

Utrecht University

The Fractional Harmonic Oscillator: Analysis and Computations

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Abstract

In this thesis, we study the behavior of the fractional harmonic oscillator using both analytical and numerical methods. Starting from the classical harmonic oscillator, we extend the model by introducing fractional derivatives, which account for memory and non-local effects.

We expand on the necessary mathematical tools such as the Gamma and Mittag-Leffler functions, and how to construct solutions to fractional differential equations. Analytical techniques such as the Laplace transform and fractional power series are used to solve linear systems, whereas numerical methods such as the L1 and L2 schemes help us approximate more cases. We apply these results to different types of simple, non-linear, damped, and driven oscillators, extending the classical approach to fractional versions of oscillating systems.

At the end, we take the small steps in extending the approach to fractional versions of other dynamical systems, like the Van der Pol and Lotka–Volterra equations. These examples show that fractional models offer new insights into systems, leaving room for further exploration.

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Chapter 1

Introduction

The birth of mathematical oscillatory motions can be traced back to our own humble nation. It was Christiaan Huygens and his outstanding work *Horologium Oscillatorium* (1673) [4, 1] who constructed the first mathematical formulation of the pendulum. It was this foundation that laid the basis for the study of harmonic motion. The tremors of his work and insights into isochronous oscillations influenced Isaac Newton in *Philosophiæ Naturalis Principia Mathematica* (1687) [12, 1]. It was, after all, with the understanding of mathematics and Newton's laws that the formalization of the classical harmonic oscillator equation came into existence [1]. This harmonic oscillator later became a cornerstone in mechanics, electromagnetism, and quantum mechanics.

But this was not the end of our story. While Isaac Newton worked on his idea of "fluxions" in *Philosophiæ Naturalis Principia Mathematica* [12], there was another lead, a lead that developed the same concept of derivatives: Gottfried Wilhelm Leibniz. In his creation *Nova Methodus pro Maximis et Minimis, Itemque Tangentibus, Qua Nec Fractas, Nec Irrationales Quantitates Moratur, & Singulare pro illis Calculi Genus* (1684) [8], he describes a new method for maxima, minima, and tangents. It is here that he tried to capsize what we nowadays call the beginning of derivatives.

Many years later, the ripples of his discovery brewed a famous exchange between *Gottfried Wilhelm Leibniz* and *Guillaume de l'Hôpital* (1695) [9]. It is in this exchange that Leibniz was curious about the meaning of a derivative of *arbitrary order*. Although it was a remark based on curiosity and not of much consequence, it sparked the foundations of fractional calculus. It was on these exchanges that *Liouville* (1832) and *Riemann* (1867) [15] formulated derivatives for non-integer differentiation, which led to the development of fractional differential equations.

As a fairly new subject in mathematics, it is still explored in this generation by *Caputo* in his paper *Linear Models of Dissipation whose Q is Almost Frequency Independent-II* (1967) [3], and *Mandelbrot*, who laid the framework for real-world phenomena in *Fractional Brownian Motions, Fractional Noises and Applications* (1968) [11].

This historical development from classical harmonic motion to the foundations of fractional calculus raises an important question: how do these fractional tools change our understanding of oscillatory systems? This thesis aims to investigate precisely that by asking: How can fractional differential equations be used to model the behavior of the fractional harmonic oscillator, and what are the implications of these models with respect to accuracy, stability, and physical interpretation?

Research question

The central research question of this thesis is:

> *How can fractional differential equations be used to model the behavior of the fractional harmonic oscillator, and what are the implications of these models with respect to accuracy, stability, and physical interpretation?*

This question guides the analytical and numerical study throughout the thesis, spanning from foundational fractional calculus to the simulation of fractional systems.

Chapter 2

Overview classical harmonic oscillator

The classical harmonic oscillator is a fundamental concept in physics and mathematics. The oscillator describes a system that tries to capture the force needed to displace an object and the displacement itself. In this chapter we will give an overview of different Harmonic Oscillators and their properties.

2.1. Simple harmonic oscillator

Definition 2.1 (Simple Harmonic Oscillator)**.** The motion of a simple harmonic oscillator follows Newton's laws of motion and is given by:

$$m\frac{d^2x}{dt^2} + kx = 0, \quad x(0) = x_0, \quad x'(0) = x_1. \tag{2.1}$$

where m is the mass of the oscillator, k is the spring constant, and x represents the displacement from the equilibrium position [6].

Lemma 2.2 (Standard form of the harmonic oscillator)**.** *The equation of motion can be rewritten in its standard form as:*

$$\frac{d^2x}{dt^2} + \omega^2 x = 0, \quad x(0) = x_0, \quad x'(0) = x_1. \tag{2.2}$$

Where $\omega = \sqrt{k/m}$ is the natural frequency of the system.

Theorem 2.3 (General solution of the simple harmonic oscillator)**.** *The general solution to the simple harmonic oscillator equation is:*

$$x(t) = x_0\cos(\omega t) + \frac{x_1}{\omega}\sin(\omega t), \tag{2.3}$$

Alternatively we can write this solution in the form:

$$x(t) = C\cos(\omega t + \phi),$$

where $C = \sqrt{(x_0)^2 + (\frac{x_1}{\omega})^2}$ is the amplitude and $\phi = -\tan^{-1}\left(\frac{\frac{x_1}{\omega}}{x_0}\right)$ is the phase shift.

Remark 2.4 (Uniqueness and existence)*.* If we have a second-order linear differential equation (2.2) with initial conditions $x(0) = x_0$ and $x'(0) = x_0$, the solution is unique and exists for all $t \in \mathbb{R}$. This follows directly from standard existence and uniqueness theorems for linear ODEs with continuous coefficients.

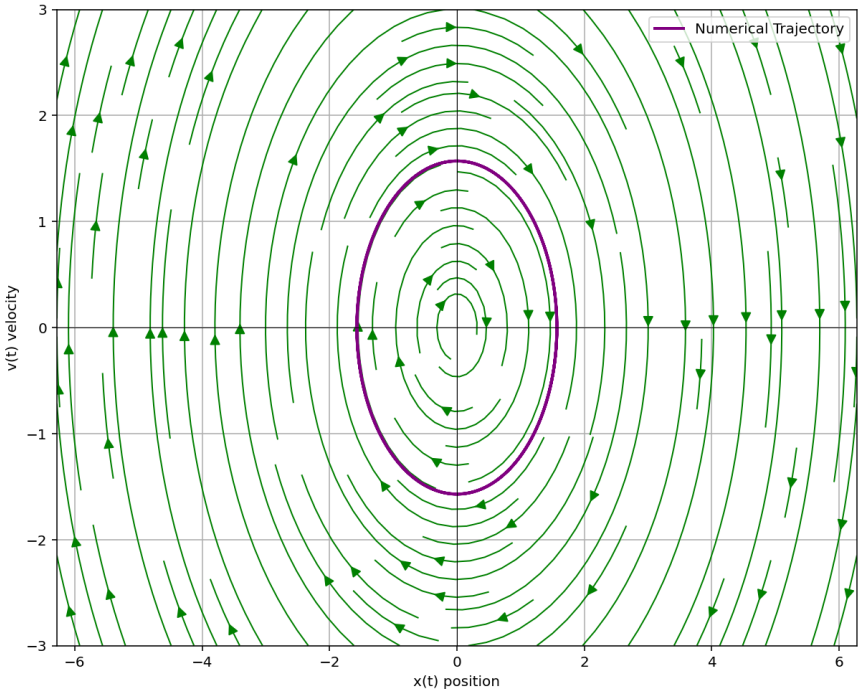


Figure 2.1: Phase portrait of the simple harmonic oscillator with $\omega^2 = 1$.

2.1.1. Phase portrait of the simple oscillator

The phase portrait represents the system in phase space, where the axes correspond to position x and velocity $v = \frac{dx}{dt}$. This leads to the system of equations,

$$\begin{cases} \frac{dx}{dt} &= v, \\ \frac{dv}{dt} &= -\omega^2 x. \end{cases} \tag{2.4}$$

Remark 2.5 (Closed orbits and frequency effects in phase space)*.* The analytical solution in definition 2.1 confirms that the phase portrait consists of closed elliptical trajectories, indicating conservation of energy. Including different values of ω in the plot could show variations in oscillatory behavior.

2.2. Non-linear oscillator

Definition 2.6 (Non-linear oscillator)**.** In many real-world systems, oscillatory motion is non-linear. If the angle of a pendulum's swing is large, the system follows:

$$\frac{d^2x}{dt^2} + \sin(x) = 0, \quad x(0) = x_0, \quad x'(0) = x_1, \tag{2.5}$$

where the restoring force is a sine function instead of a linear function [6, p. 17] .

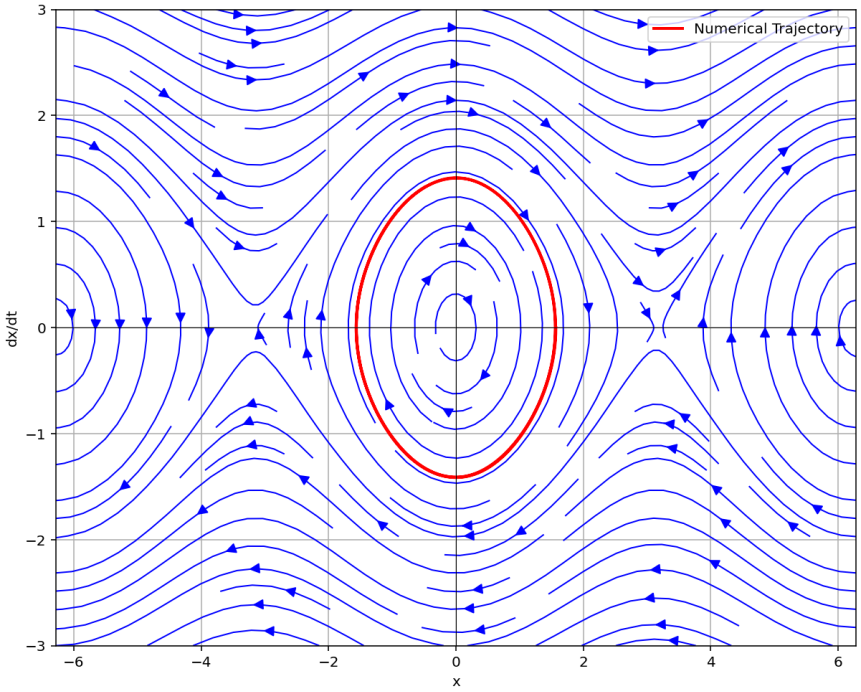


Figure 2.2: Phase portrait of the Simple Harmonic Oscillator with $\omega^2 = 1$.

Theorem 2.7 (Solution for the non-linear oscillator)**.** *Unlike the simple harmonic oscillator, the non-linear equation does not have a closed-form solution in elementary functions. Instead, its solution is given by the integral:*

$$\int \frac{dx}{\sqrt{2(C + \cos(x))}} = t. \tag{2.6}$$

2.2.1. Phase portrait of the non-linear oscillator

Remark 2.8 (Non-Elliptical Trajectories in the Non-linear Oscillator)*.* In contrast to the linear oscillator 2.2, the trajectories in the phase space are no longer perfectly elliptical. They are slightly expanded for larger amplitudes compared to the linear restoring force. As the amplitude increases, the trajectories become more stretched out.

2.3. Damped harmonic oscillator

Definition 2.9 (Damped harmonic oscillator)**.** In real-world oscillators, resistance from friction or air damping reduces oscillation amplitude. The equation of motion for a damped harmonic oscillator is:

$$m\frac{d^2x}{dt^2} + b\frac{dx}{dt} + kx = 0, \quad x(0) = x_0, \quad x'(0) = x_1 \tag{2.7}$$

where b is the damping coefficient [6].

Theorem 2.10 (General solution damped harmonic oscillator)**.** *The general solution is given by:*

$$x(t) = e^{-\alpha t}\left(C_1 e^{(\sqrt{\omega^2-\alpha^2})t} + C_2 e^{-(\sqrt{\alpha^2-\omega^2})t}\right), \tag{2.8}$$

where $\alpha = \frac{b}{2m}$ *and* $\omega = \sqrt{\frac{k}{m}}$*.*

Remark 2.11 (Classification of damping based on ω and α)*.* The general solution contains three distinct cases depending on the relationship between α and ω: the underdamped case ($\omega^2 > \alpha^2$), the critically damped case ($\omega^2 = \alpha^2$), and the overdamped case ($\omega^2 < \alpha^2$). These cases influence the behavior of the oscillator such that the motion is either oscillatory, non-oscillatory, or rapidly damped.

Example 2.12 (Underdamped case: $\omega^2 > \alpha^2$)**.** In the underdamped case, the solution exhibits oscillations with amplitude decreasing exponentially:

$$x(t) = e^{-\alpha t}(C_1 \cos(\gamma t) + C_2 \sin(\gamma t)), \tag{2.9}$$

where $\gamma = \sqrt{\omega^2 - \alpha^2}$.

Remark 2.13*.* In the underdamped scenario, the system returns to equilibrium through oscillations with gradually diminishing amplitudes.

Example 2.14 (Critically damped case: $\omega^2 = \alpha^2$)**.** In the critically damped case, the system returns to equilibrium in the shortest possible time without oscillating. The solution is given by:

$$x(t) = (C_1 + C_2 t)e^{-\alpha t}. \tag{2.10}$$

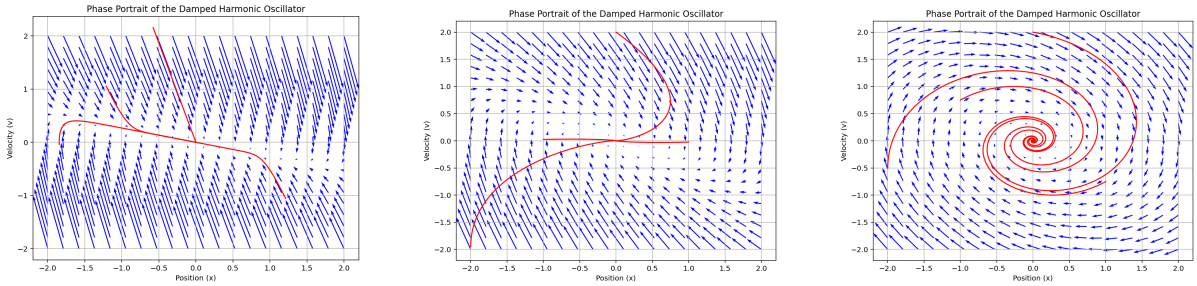
Remark 2.15 (Threshold for decay and oscillation)*.* Critical damping represents the boundary condition between oscillatory and non-oscillatory behavior, returning quickly to equilibrium.

Example 2.16 (Overdamped case: $\omega^2 < \alpha^2$)**.** In the overdamped case, the solution is characterized by a slow return to equilibrium without oscillations:

$$x(t) = e^{-\alpha t}(C_1 e^{\gamma t} + C_2 e^{-\gamma t}), \tag{2.11}$$

where $\gamma = \sqrt{\alpha^2 - \omega^2}$.

Remark 2.17 (Behavior of overdamped systems)*.* The overdamped scenario reflects highly damped systems, gradually approaching equilibrium without oscillatory motion.



(a) Overdamped (b) Critically damped (c) Underdamped

Figure 2.3: Phase portraits of the damped harmonic oscillator for three damping regimes.

2.3.1. Phase portrait of the harmonic oscillator

Remark 2.18. The phase portrait of the damped harmonic oscillator differs significantly from that of the simple harmonic oscillator. The damping parameter influences whether the system exhibits underdamping, critical damping, or overdamping behavior.

Remark 2.19 (Phase portraits). The phase portrait of the underdamped oscillator spirals inward to equilibrium; the critically damped oscillator approaches equilibrium quickly and directly, while the overdamped oscillator approaches equilibrium slowly without spiraling.

2.4. Driven harmonic oscillator

Definition 2.20 (Driven harmonic oscillator)**.** In many real-world applications, oscillators are subjected to external forces. The equation of motion for a driven harmonic oscillator is:

$$m\frac{d^2x}{dt^2} + b\frac{dx}{dt} + kx = F(t), \tag{2.12}$$

where $m > 0$ is the mass, $k > 0$ is the spring constant, $b \geq 0$ is the damping coefficient, $F(t)$ is the driving force.

Lemma 2.21 (General form of solution)**.** *The general solution to the driven harmonic oscillator* (2.12) *is composed of two parts:*

$$x(t) = x_h(t) + x_p(t), \tag{2.13}$$

where $x_h(t)$ is the homogeneous solution and $x_p(t)$ the particular solution.

Remark 2.22 (On the form of the homogeneous solution)*.* The homogeneous solution $x_h(t)$ corresponds to the damped harmonic oscillator equation given previously by equation (2.8), meaning that the form of $x_h(t)$ depends on whether the system is under-damped, critically damped, or over-damped.

Theorem 2.23 (Particular solution for periodic driving force)**.** *If we consider a driving force of the periodic form:*

$$F(t) = F_0 \cos(\omega_d t), \tag{2.14}$$

where F_0 is the amplitude of the driving force and ω_d is its frequency, then the particular solution of equation (2.12) *takes the form:*

$$x_p(t) = A\cos(\omega_d t) + B\sin(\omega_d t), \tag{2.15}$$

where coefficients A and B are given explicitly by:

$$C_1 = \frac{(k - m\omega_d^2)F_0}{(k - m\omega_d^2)^2 + (b\omega_d)^2}, \quad C_2 = \frac{b\omega_d F_0}{(k - m\omega_d^2)^2 + (b\omega_d)^2}. \tag{2.16}$$

Remark 2.24*.* The homogeneous solution reflects the natural motion of the system, while the particular solution expresses the influence of the driving force.

2.4.1. Resonance of the driven oscillator

Remark 2.25 (Resonance)*.* When the driving frequency ω_d approaches the natural frequency of the system, $\omega = \sqrt{k/m}$, we observe a phenomenon called resonance.

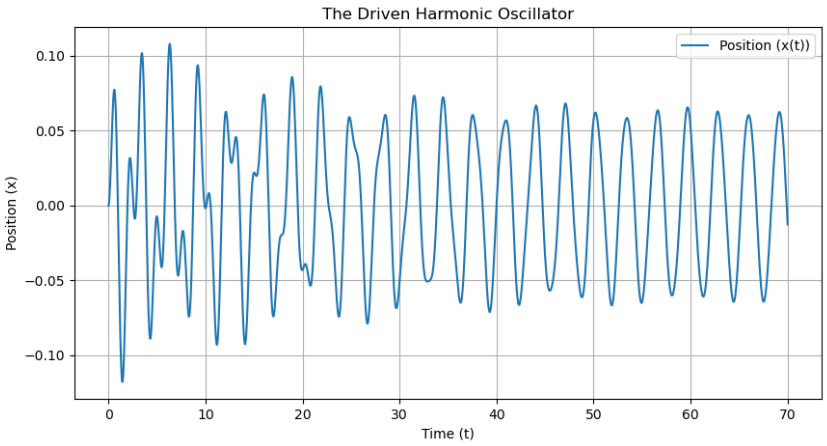


Figure 2.4: The driven harmonic oscillator.

Remark 2.26 (Resonance and long term behavior)*.* We notice the influence of the driving force on the natural solution during the early time period. After some time, due to the resonance of these waves, the system returns to a simple oscillating motion.

Chapter 3

Overview fractional calculus

In this chapter, we introduce the most commonly used definitions of fractional derivatives, such as the Grünwald-Letnikov, Riemann-Liouville, and Caputo derivatives. We examine their mathematical properties such as linearity and commutativity, and reduction to classical derivatives. We also inspect the conditions under which these derivatives coincide. In addition, we introduce analytical tools like the Laplace transform and show their role in solving fractional differential equations. Lastly, we conclude with a short discussion on equilibrium and stability analysis within respect to the fractional models.

3.1. Mathematical Tools for Fractional Calculus

Before we can view at the fractional derivatives and their application. We need to extend our toolbox to handle these concepts. In this section we will look at the definitions and theorems needed to explore fractional derivatives.

3.1.1. Gamma function

Definition 3.1 (Gamma function [7, p. 24])**.** The Gamma function $\Gamma(x)$ is an extension of the factorial function from integers to all positive real numbers:

$$\Gamma(x) = \int_0^\infty t^{x-1} e^{-t} dt \qquad x \in \mathbb{R}^+. \tag{3.1}$$

The integral is absolutely convergent for all real values $x > 0$, and the Gamma function is a smooth function on $\mathbb{R}^+$.

Remark 3.2 (Relation to the factorial function)*.* As the factorial function of n is built on the multiplication of all positive integers less than or equal to n, the Gamma function is based on the same concept. When n is a positive integer, it coincides with the factorial function. For positive real values that are not integers, we can make use of the recurrence property.

Theorem 3.3 (Recurrence Gamma function [7, p. 24])**.** *The recurrence property of the gamma function is the relation:*

$$\Gamma(x+1) = x\Gamma(x). \tag{3.2}$$

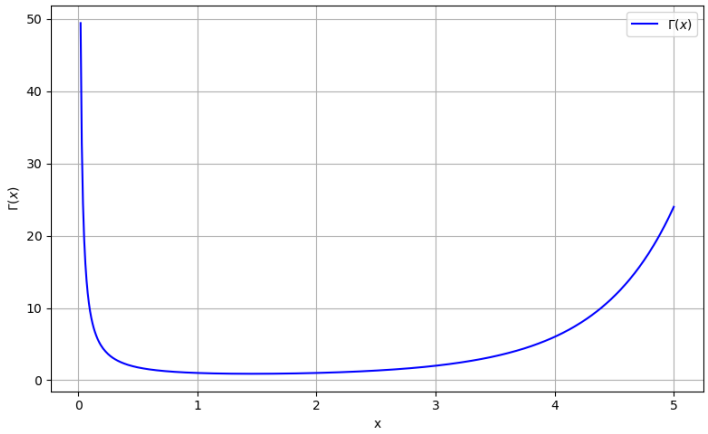


Figure 3.1: Plot of the Gamma function $\Gamma(x)$ for $x > 0$.

Theorem 3.4 (Relation Gamma function and factorials)**.** *The Gamma function is an extension of the factorial. For positive integers, we have:*

$$\Gamma(n) = n! \qquad n \in \mathbb{N}. \tag{3.3}$$

Lemma 3.5 (Gamma function and Stirling's [5, p. 20])**.** *Let $x > 0$ and suppose $x \to \infty$. Then the Gamma function admits the following asymptotic approximation:*

$$\Gamma(x) \sim \sqrt{2\pi}\, x^{x-\frac{1}{2}} e^{-x} \left(1 + O\left(x^{-1}\right)\right). \tag{3.4}$$

This shows that the Gamma function grows faster than exponentially.

Remark 3.6 (Link to binomial coefficients)*.* Now that we have defined an extension of the factorial function to positive real numbers and developed an understanding of its behavior, we can link the Gamma function to the binomial coefficients in the same way that factorials are linked to classical binomial coefficients.

Definition 3.7 (Binomial coefficients for real values)**.** The binomial coefficient for real-valued arguments is defined by:

$$\binom{\alpha}{\beta} = \frac{\Gamma(\alpha+1)}{\Gamma(\alpha-\beta+1)\Gamma(\beta+1)} \qquad \alpha, \beta \in \mathbb{R}. \tag{3.5}$$

Remark 3.8 (Extension of the binomial series to real exponents)*.* This definition also extends the classical binomial formula beyond integers, allowing us to give mathematical context to the binomial series for real exponents:

$$(1+x)^{\alpha} = \sum_{k=0}^{\infty} \binom{\alpha}{k} x^k, \quad \text{for } |x| < 1,\ \alpha \in \mathbb{R}. \tag{3.6}$$

With this, we can also extend many other series beyond the scope of positive integers, such as the exponential function, which is defined in terms of the factorial.

3.1.2. Mittag-Leffler function

Definition 3.9 (Mittag-Leffler function[14, p.17])**.** The Mittag-Leffler function $E_{\alpha,\beta}(z)$ is a natural extension of the power series of the exponential function, defined with respect to the Gamma function as:

$$E_{\alpha,\beta}(z) = \sum_{n=0}^{\infty} \frac{z^n}{\Gamma(\alpha n + \beta)} \qquad Re(\alpha) > 0, \tag{3.7}$$

where this function is composed of the complex valued $\alpha, \beta \in \mathbb{R}$ and $z \in \mathbb{C}$.

Remark 3.10 (Convergence of the Mittag-Leffler function series)*.* Let $\alpha > 0$ and $\alpha, \beta \in \mathbb{R}$. Then the Mittag-Leffler function $E_{\alpha,\beta}(z)$ converges absolutely for all $z \in \mathbb{R}$. From this, we conclude that $E_{\alpha,\beta}(z)$ defines a continuous and infinitely differentiable function on $\mathbb{R}$.

This convergence follows from the fact that the denominator $\Gamma(\alpha n+\beta)$ grows faster than exponentially as $n \to \infty$ (see Lemma 3.5).

Theorem 3.11 (Derivative of the two-Parameter Mittag-Leffler function [7, p. 44])**.** *Let $\alpha > 0$, $\beta \in \mathbb{R}$, and $n \in \mathbb{N}$. Then the n-th derivative of the two-parameter Mittag-Leffler function with respect to its argument z is given by:*

$$\frac{d^n}{dz^n} E_{\alpha,\beta}(z) = E_{\alpha,\beta-n}(z). \tag{3.8}$$

Remark 3.12 (Mittag-Leffler and differential equations)*.* The Mittag-Leffler function plays a central role in solving fractional differential equations and will appear frequently in later sections.

Remark 3.13 (Exponential function and Mittag-Leffler [14, p. 18])*.* The Mittag-Leffler function generalization the exponential function. In the special case when $\alpha = 1$ and $\beta = 1$, we recover the classical exponential series:

$$E_{1,1}(z) = \sum_{k=0}^{\infty} \frac{z^k}{\Gamma(k+1)} = \sum_{k=0}^{\infty} \frac{z^k}{k!} = e^z. \tag{3.9}$$

Lemma 3.14 (Trigonometric-functions as Mittag-Leffler function [14, p. 19].)**.** *Let $\omega > 0$. Then the cosine and sin function can be expressed as a special case of the Mittag-Leffler function:*

$$E_{2,1}(-\omega^2 z^2) = \cos(\omega z). \tag{3.10}$$

Remark 3.15 (Connection to classical trigonometric functions). These identities hold when the order $\alpha = 2$ and demonstrate how classical trigonometric functions, such as cosine, are connected by the Mittag-Leffler function.

Theorem 3.16 (Asymptotic growth of Mittag-Leffler function for positive Values [14, p. 34]). *Let $\alpha > 0$, $\beta \in \mathbb{R}$, and $z > 0$. Then, as $z \to \infty$, the two-parameter Mittag-Leffler function satisfies the asymptotic behavior:*

$$E_{\alpha,\beta}(z) \sim \frac{1}{\alpha} \exp\left(z^{1/\alpha}\right) z^{(1-\beta)/\alpha}, \quad z \to \infty. \tag{3.11}$$

This shows that, for large positive values of z, the Mittag-Leffler function has some form of super-exponential growth. As the value of α decreases, the growth becomes stronger.

Theorem 3.17 (Asymptotic decay of the Mittag-Leffler function negative values [14, p. 35]). *Let $0 < \alpha < 2$ and $\beta \in \mathbb{R}$, with $z > 0$. Then the Mittag-Leffler function satisfies the following asymptotic behavior as $z \to \infty$:*

$$E_{\alpha,\beta}(-z) \sim \sum_{k=1}^{N} \frac{(-1)^{k+1}}{z^k \Gamma(\beta - \alpha k)} + \mathcal{O}\left(z^{-N-1}\right), \quad z \to \infty. \tag{3.12}$$

where N is a truncation index, which can be chosen depending on the desired number of terms and accuracy of the approximation.

Now that we have established the foundation on the special functions, especially the Gamma and Mittag-Leffler functions. We now turn to the core definitions of fractional calculus.

3.2. Fractional derivatives and integrals

Fractional calculus has many different definitions for non-integer order derivatives. Each of these definition has their own strengths and context. We will discuss the most commonly used definition with respect to computation and numerical branch and discuss their strengths.

3.2.1. Fractional derivative

Starting with the Grünwald-Letnikov derivative. Grünwald-Letnikov provides an extension of the classical derivative in terms of a limit-based approach. We look at a generalization of the finite difference method extended to real numbers [13, p.15].

Definition 3.18 (Grünwald-Letnikov fractional derivative [14, p. 52])**.** Let $\alpha > 0$ and let $f(t)$ be defined on the interval $[a, t]$. The Grünwald–Letnikov fractional derivative of order α is defined as:

$$ {}^{GL}_{a}D^{\alpha}_{t} f(t) = \lim_{h \to 0} \frac{1}{h^{\alpha}} \sum_{k=0}^{\left\lfloor \frac{t-a}{h} \right\rfloor} (-1)^k \binom{\alpha}{k} f(t - kh), \tag{3.13} $$

where $\binom{\alpha}{k}$ is the generalized binomial coefficient as defined in Equation (3.5). This definition requires that f is at least n times continuously differentiable on $[a, t]$, where $n = \lceil \alpha \rceil$.

Remark 3.19 (Connection to classical difference quotient)*.* The Grünwald-Letnikov derivative is directly connected to the definition of the classical difference quotient of the derivative, making it an intuitive extension [13, p.16].

Definition 3.20 (Riemann-Liouville fractional derivative [14, p. 69])**.** Let $\alpha > 0$ with $n = \lceil \alpha \rceil$, and let $f(t)$ be defined on the interval $[a, t]$. Suppose that $f \in L^1[a, t]$ and that f is at least n times continuously differentiable on $[a, t]$. The left-sided Riemann-Liouville fractional derivative of order α is defined as:

$$ {}^{RL}_{a}D^{\alpha}_{t} f(t) = \frac{1}{\Gamma(n - \alpha)} \frac{d^n}{dt^n} \int_a^t f(\tau)(t - \tau)^{n-\alpha-1} \, d\tau, \tag{3.14} $$

where $\Gamma(\cdot)$ is the Gamma function 3.1.1..

Remark 3.21 (Integral based approach)*.* Unlike the Grünwald-Letnikov definition, where we saw an approach based on the finite difference method, the Riemann–Liouville derivative takes an integral based approach to fractional differentiation.

Remark 3.22 (Grünwald-Letnikov and Riemann-Liouville relation)*.* Even though the Grünwald–Letnikov and Riemann–Liouville fractional derivatives are defined using different approaches. Under certain conditions, the two definitions coincide and give equivalent results, showing the unification within fractional calculus.

Theorem 3.23 (Equivalence of Grünwald-Letnikov and Riemann-Liouville derivatives [14, p. 75-77])**.** *Let $\alpha > 0$ and let $n = \lceil \alpha \rceil$. Suppose that $f \in C^{n-1}([a, t])$ and that $f^{(n)} \in L^1([a, t])$. Then the Grünwald–Letnikov and Riemann–Liouville fractional derivatives of order α are equivalent:*

$$ {}^{GL}_{a}D^{\alpha}_{t} f(t) = {}^{RL}_{a}D^{\alpha}_{t} f(t). \tag{3.15} $$

Remark 3.24 (Importance of the equivalence theorem)*.* The equivalence between the Grünwald–Letnikov and Riemann–Liouville fractional derivatives is important because it confirms a unifying framework within fractional calculus. It makes sure, under suitable smoothness and integrability conditions, different approaches to defining fractional derivatives lead to the same analytical results, bridging the formulations.

Definition 3.25 (Caputo fractional derivative [14, p. 91])**.** Let $\alpha > 0$ and let $n = \lceil \alpha \rceil$. Suppose $f \in C^n([a,t])$. Then the left-sided Caputo fractional derivative of order α is defined by:

$$ {}_a^C D_t^\alpha f(t) = \frac{1}{\Gamma(n-\alpha)} \int_a^t \frac{f^{(n)}(\tau)}{(t-\tau)^{\alpha-n+1}}\, d\tau, \tag{3.16} $$

where $\Gamma(\cdot)$ is the Gamma function, and $f^{(n)}$ denotes the n-th classical derivative of f.

Remark 3.26 (Initial conditions: Caputo vs. Riemann–Liouville)*.* While both the Riemann-Liouville and Caputo derivatives are defined using integral operators, they differ in how they handle initial conditions. The Riemann-Liouville derivative requires initial conditions expressed in terms of fractional integrals, which are non-intuitive and hard to get. The Caputo derivative has initial conditions in terms of integer-order derivatives, making it more suitable for modelling and physical applications.

Theorem 3.27 (Equivalence of Caputo and Riemann-Liouville derivatives [14, p. 85])**.** *Let $\alpha > 0$ and let $n = \lceil \alpha \rceil$. Suppose $f \in C^{n-1}([a,b])$ and $f^{(n)} \in L^1([a,b])$. If the initial conditions:*

$$ f^{(k)}(a) = 0 \quad \textit{for all } \ 0 \le k < n. \tag{3.17} $$

Then the Caputo and Riemann-Liouville fractional derivatives of order α are equivalent:

$$ {}_a^C D_t^\alpha f(t) = {}_a^{RL} D_t^\alpha f(t), \quad \textit{for all } t \in (a,b). \tag{3.18} $$

With the important definitions defined, we can now examine which properties, such as linearity and commutativity, extend to fractional derivatives.

3.2.2. Properties Fractional Derivatives

Remark 3.28 (Linearity of fractional derivatives)*.* Linearity is one of the few classical properties that extends directly to all major definitions of fractional derivatives. This makes it particularly valuable for both analytical manipulations and numerical schemes.

Theorem 3.29 (Linearity of fractional derivatives [14, p. 90])**.** *Let $\alpha > 0$, and let $f(t)$ and $g(t)$ be functions for which the respective fractional derivatives exist in some coinciding interval $t \in [a, b]$. Then the Riemann–Liouville, Grünwald–Letnikov, and Caputo fractional derivatives are linear operators. For all constants $\lambda, \mu \in \mathbb{R}$, we have:*

$$\begin{aligned}
{}^{RL}_{a}D^{\alpha}_{t}\left[\lambda f(t) + \mu g(t)\right] &= \lambda\, {}^{RL}_{a}D^{\alpha}_{t} f(t) + \mu\, {}^{RL}_{a}D^{\alpha}_{t} g(t),\\
{}^{GL}_{a}D^{\alpha}_{t}\left[\lambda f(t) + \mu g(t)\right] &= \lambda\, {}^{GL}_{a}D^{\alpha}_{t} f(t) + \mu\, {}^{GL}_{a}D^{\alpha}_{t} g(t),\\
{}^{C}_{a}D^{\alpha}_{t}\left[\lambda f(t) + \mu g(t)\right] &= \lambda\, {}^{C}_{a}D^{\alpha}_{t} f(t) + \mu\, {}^{C}_{a}D^{\alpha}_{t} g(t).
\end{aligned}$$

Remark 3.30 (Linearity inherited from classical operators)*.* This property follows directly from the linearity of the underlying integral and differential operators used in the definitions of the fractional derivatives.

Theorem 3.31 (Reduction to classical derivatives)**.** *Let $n \in \mathbb{N}$ and suppose $f \in C^n([a, b])$. Then for all $t \in (a, b)$, the fractional derivatives of integer order $\alpha = n$ reduce to the classical n-th derivative:*

$$\begin{aligned}
{}^{RL}_{a}D^{n}_{t} f(t) &= \frac{d^n}{dt^n} f(t),\\
{}^{GL}_{a}D^{n}_{t} f(t) &= \frac{d^n}{dt^n} f(t),\\
{}^{C}_{a}D^{n}_{t} f(t) &= \frac{d^n}{dt^n} f(t).
\end{aligned}$$

Remark 3.32 (Validation against classical calculus)*.* This property is not only theoretically important as a confirmation that fractional derivatives generalize classical derivatives [Validation Against Classical Calculus]but also practically useful. It enables the validation of analytical and numerical methods by checking that they recover classical behavior when the order of differentiation is an integer.

Theorem 3.33 (Fractional derivative of a constant function)**.** *Let $c \in \mathbb{R}$ be a constant and $\alpha > 0$. Then the fractional derivatives of c are given by:*

$$\begin{aligned}
{}^{C}_{a}D^{\alpha}_{t} c &= 0,\\
{}^{RL}_{a}D^{\alpha}_{t} c &= \frac{c}{\Gamma(1-\alpha)}(t-a)^{-\alpha},\\
{}^{GL}_{a}D^{\alpha}_{t} c &= \frac{c}{\Gamma(1-\alpha)}(t-a)^{-\alpha}.
\end{aligned}$$

Theorem 3.34 (Non-commutativity of fractional derivatives [14, p. 71])**.** *Let $\alpha, \beta > 0$, and let D^{α} and D^{β} denote fractional derivatives of the same type either Riemann–Liouville, Grünwald–Letnikov, or Caputo. In general, the composition of fractional derivatives is not commutative:*

$$D^{\alpha} D^{\beta} f(t) \neq D^{\beta} D^{\alpha} f(t). \tag{3.19}$$

This inequality holds unless the function $f(t)$ satisfies additional conditions stated in Podlubny's Fractional Differential Equations.

Remark 3.35 (Interpretation of non-commutativity)*.* This theorem shows a big difference between classical calculus and fractional calculus. While integer-order derivatives commute under mild assumptions, fractional derivatives require much stricter conditions. This makes it that we must be careful considering the analytical manipulations and numerical implementations involving operator compositions.

Remark 3.36 (Memory effect fractional derivatives)*.* Fractional derivatives are inherently non-local due to their integral definitions, such as 3.20 and 3.25.This non-locality induces a so called memory effect, meaning that the value of the derivative at time t depends on the entire history of the function up to that point.

Remark 3.37 (Non-periodicity of fractional derivatives [16])*.* Unlike integer-order derivatives, fractional derivatives do not preserve periodicity. Even if we have a periodic function $f(t)$, its fractional derivative $D^\alpha f(t)$ (for $0 < \alpha < 1$ or any $\alpha \notin \mathbb{N}$) is generally not periodic. This causes that the existence of periodic solutions in autonomous fractional-order systems like the fractional oscillators (see in chapter 6) to be non-existing.

3.2.3. Fractional derivatives of standard functions

In classical calculus the derivatives of elementary functions such as the power function, exponential function, and trigonometric functions are well known and play a central role in solving differential equations. In fractional calculus some of these functions continue to be very important, but their fractional derivatives have different forms and are not always intuitive. In this subsection, we will discuss the fractional derivatives of the power function and the mittag-leffler functions.

Theorem 3.38 (Riemann-Liouville Derivative of a Mittag-Leffler function [14, p.])**.** *Let $\alpha > 0$, $\beta > 0$, $\lambda \in \mathbb{R}$, and $0 < \mu < \beta$. Then we defined the Riemann–Liouville fractional derivative of the Mittag-Leffler function is given by:*

$$ {}^{RL}_{a}D^{\mu}_{t}[(t-a)^{\beta-1}E_{\alpha,\beta}(\lambda(t-a)^{\alpha})] = t^{\beta-\mu-1}E_{\alpha,\beta-\mu}(\lambda(t-a)^{\alpha}), \quad t \in (a,\infty) \tag{3.20} $$

Remark 3.39 (Eigenfunction property for special case)*.* When the two-parameter Mittag-Leffler function $E_{\alpha,\beta}(\lambda t^{\mu})$ satisfies $\beta = 1$ and $\mu = \alpha$, we obtain the special case:

$$ {}^{GL}_{a}D^{\alpha}_{t}\left[E_{\alpha,1}(\lambda(t-a)^{\alpha})\right] = \lambda E_{\alpha,1}(\lambda(t-a)^{\alpha}). $$

This special case confirms that the function $E_{\alpha,1}(\lambda(t-a)^{\alpha})$ is an eigenfunction α, with eigenvalue λ. The implications of this will be discussed in section 4.2. where we link linear fractional differential equations to this property [7, p.28].

Theorem 3.40 (Riemann-Liouville derivative of a power function [14, p. 55])**.** *Let $\alpha > 0$ and $\mu > \alpha - 1$, and $t > a$. Then the Riemann-Liouville fractional derivative of the power function $(t-a)^{\mu}$ is given by,*

$$ {}^{RL}_{a}D^{\alpha}_{t}[(t-a)^{\mu}] = \frac{\Gamma(\mu+1)}{\Gamma(\mu-\alpha+1)}(t-a)^{\mu-\alpha}, \quad t \in (a,\infty) $$

Theorem 3.41 (Grünwald-Letnikov derivative of a power function [14, p. 56])**.** *Let $\alpha > 0$, $\mu > \alpha - 1$, and $t > a$. Then the Riemann–Liouville fractional derivative of the power function $(t-a)^{\mu}$ is given by:*

$$ {}^{GL}_{a}D^{\alpha}_{t}[(t-a)^{\mu}] = \frac{\Gamma(\mu+1)}{\Gamma(\mu-\alpha+1)}(t-a)^{\mu-\alpha}, \quad t \in (a,\infty). \tag{3.21} $$

Remark 3.42 (Coincidence of Grünwald-Letnikov and Riemann-Liouville derivatives of the Mittag-Leffler function)*.* The definitions of the power functions of the Grünwald-Letnikov derivative coincide the Riemann-Liouville derivative of the power function. This is in alignment of th equivalence theorem discussed in theorem 3.22, since the power function is n-times differentiable on $t \in (a,\infty)$.

Theorem 3.43. *Caputo Fractional derivative of the Mittag-Leffler function [7, p. 98]] Let $\alpha > 0$, $a \in \mathbb{R}$, and $\lambda \in \mathbb{R}$. Then the Caputo fractional derivative of the Mittag-Leffler function $E_{\alpha,1}(\lambda(t-a)^{\alpha})$ is given by:*

$$ {}^{C}_{a}D^{\alpha}_{t}[E_{\alpha,1}\big(\lambda(t-a)^{\alpha}\big)] = \lambda E_{\alpha,1}\big(\lambda(t-a)^{\alpha}\big), \tag{3.22} $$

This shows that $E_{\alpha,1}(\lambda(t-a)^{\alpha})$ is an eigenfunction of the Caputo derivative operator with eigenvalue λ.

Remark 3.44 (Coincidence of Caputo and Riemann-Liouville derivatives of the Mittag-Leffler function)*.* In the special case where the parameters satisfy $\mu = \alpha$ and $\beta = 1$, the Caputo and Riemann-Liouville fractional derivatives produces the same result when applied on the mittag-leffler function. This is one of the rare non-trivial cases where both definitions of the fractional derivative coincide without the conditions in theorem 3.27. This shows that the Mittag-Leffler function has an overarching role in fractional calculus.

3.3. Analytical tools for fractional differential equations

To analyze and solve fractional differential equations (FDEs), it is essential to understand the underlying framework. This includes the functional requirements of fractional derivatives, the structure of equations, and the types of initial conditions that must be specified. Unlike classical differential equations, FDE's exhibit memory effects and require integral operator formulations, which impose additional analytical and numerical considerations.

3.3.1. Operational framework of fractional differential equations

To analyze fractional differential equations (FDEs), we need an operational framework that extends classical theory to fractional theory. This includes conditions for existence and uniqueness, appropriate initial values, and the integral structure inherent to fractional derivatives. The following results form the foundation for well-posedness and further analytical techniques.

Theorem 3.45 (Existence and uniqueness for Riemann-Liouville initial value problems [7][p.148])**.** *Let $\alpha > 0$ and $n = \lceil \alpha \rceil$. Suppose $g(t, f)$ is continuous on $[a, b] \times \mathbb{R}$ and Lipschitz continuous in f. Consider the initial value problem:*

$$ {}_a^C D_t^\alpha f(t) = g(t, f(t)), \quad \left(I_a^{n-\alpha+k} x\right)(t)\big|_{t=a} = c_k, \quad k = 0, 1, \ldots, n-1. \tag{3.23} $$

Then there exists a unique solution $f(t) \in L^1([a, b])$, given that the initial conditions $\{c_k\}$ are well-defined and finite. We define $I_a^{n-\alpha+k} f)(t)$ as the Riemann-Liouville fractional integral:

$$ (I_a^{n-\alpha+k} f)(t) = \frac{1}{\Gamma(n-\alpha+k)} \int_a^t (t-\tau)^{n-\alpha+k-1} f(\tau)\, d\tau. $$

Remark 3.46 (Limitations of Riemann–Liouville initial conditions)*.* These conditions are generally non-local and limitations of Riemann–Liouville Initial Conditions are no direct physical interpretation, which may limit the usage of the Riemann-Liouville formulation in problems with explicit initial conditions stated.

Theorem 3.47 (Existence and uniqueness for Caputo initial value problems [7][p.199-205])**.** *Let $\alpha > 0$ and $n = \lceil \alpha \rceil$. Suppose $g(t, f)$ is continuous on $[a, b] \times \mathbb{R}$ and Lipschitz continuous in f. Consider the initial value problem:*

$$ {}_a^C D_t^\alpha f(t) = g(t, f(t)), \quad u^{(k)}(a) = u_k, \quad k = 0, 1, \ldots, n-1. \tag{3.24} $$

If g satisfies the Lipschitz condition and f is uniform on the compact subsets of t, then there exists a unique solution $f(t) \in C^n([a, b])$.

Remark 3.48 (On the nature of initial conditions)*.* The difference between the Caputo and Riemann-Liouville formulation of the problems lies in the type of initial conditions they have. Where Caputo derivatives are designed to have more in classical sense initial conditions involving integer-order derivatives, which are sometimes measurable depending on the physical interpretations. Is the Riemann-Liouville designed for initial conditions in terms of fractional integrals, which are generally more difficult to find.

Theorem 3.49 (Existence and uniqueness for two-term Caputo fractional differential equations [7][p.153=157])**.** *Let $\alpha, \beta > 0$ with $\alpha \neq \beta$, and let $n = \lceil \max(\alpha, \beta) \rceil$.*

Suppose $p, q \in \mathbb{R}$ and $g(t, f)$ is continuous on $[a, b] \times \mathbb{R}$ and Lipschitz continuous in f. Consider the initial value problem:

$$p\ {}_a^C D_t^\alpha f(t) + q\ {}_a^C D_t^\beta f(t) = g(t, f(t)), \quad f^{(k)}(a) = c_k, \quad k = 0, 1, \ldots, n-1. \tag{3.25}$$

Then there exists a unique solution $f(t) \in C^n([a, b])$.

Theorem 3.50 (Existence and uniqueness for two-term Riemann-Liouville fractional differential equations [7] [pp. 199-205])**.** *Let $\alpha, \beta > 0$ with $\alpha \neq \beta$, and let $n = \lceil \max(\alpha, \beta) \rceil$. Suppose $p, q \in \mathbb{R}$ and $g(t, f)$ is continuous on $[a, b] \times \mathbb{R}$ and Lipschitz continuous in f. Consider the initial value problem:*

$$p\ {}_a^{RL} D_t^\alpha f(t) + q\ {}_a^C D_t^\beta f(t) = g(t, f(t)), \quad t \in (a, b]. \tag{3.26}$$

With initial conditions:

$$(I_a^{n-\alpha+k} f)(t)\Big|_{t=a} = c_k, \quad k = 0, 1, \ldots, n-1. \tag{3.27}$$

Then there exists a unique solution $f(t) \in L^1([a, b])$.

Remark 3.51 (Application to the fractional harmonic oscillator)*.* The two-term fractional differential equations discussed above, in both the Caputo and Riemann-Liouville senses, form the analytical foundation for modelling the Fractional Harmonic Oscillator. In particular, this framework will be applied later to analyze the fractional harmonic oscillator.

Remark 3.52 (On the choice of derivative)*.* The choice between Caputo and Riemann-Liouville derivatives is problem dependent. When modeling physical systems where initial data are naturally specified in terms of their classical derivatives like displacement and velocity, the Caputo derivative is generally preferred. While the Riemann-Liouville derivatives may be more appropriate in the abstract sense when integral formulations naturally comes forth.

3.3.2. Laplace transforms for fractional derivatives

The Laplace transform is a strong tool for solving linear fractional differential equations (FDEs), especially those with constant coefficients. This comes from the fact that the Laplace transformation changes the fractional derivative of a function into a complex algebraic equation. From which we will later find explicit solutions in 4.3..

Definition 3.53 (Laplace transform and conditions on existence)**.** Let $f : [0, \infty) \to \mathbb{R}$ be a piecewise continuous function on every finite interval $[0, T]$ for $T > 0$. The Laplace transform of f is defined by:

$$\mathcal{L}\{f(t)\}(s) = \int_0^\infty e^{-st} f(t)\, dt, \quad s \in \mathbb{C}. \tag{3.28}$$

The Laplace transform $\mathcal{L}\{f(t)\}(s) = \tilde{f}(s)$ exists if there exist constants $M > 0$ and $\omega \in \mathbb{R}$ such that:

$$|f(t)| \leq Me^{\omega t} \quad \text{for all } t \geq 0. \tag{3.29}$$

Remark 3.54 (On the existence of the inverse Laplace transform [14, p. 103])*.* The inverse Laplace transform exists when the Laplace transformed function $F(s)$ is analytic in the positive real plane and it satisfies the standard conditions for the Laplace transformation.

Theorem 3.55 (Linearity of the Laplace transform and its inverse [7, p. 18])**.** *First $f(t)$ and $g(t)$ be functions for which the Laplace transforms $\mathcal{L}\{f(t)\} = F(s)$ and $\mathcal{L}\{g(t)\} = G(s)$ exist in a common region of convergence. Then for all constants $a, b \in \mathbb{C}$, the Laplace transform and its inverse are linear operators:*

$$\mathcal{L}\{af(t) + bg(t)\} = aF(s) + bG(s), \tag{3.30}$$
$$\mathcal{L}^{-1}\{aF(s) + bG(s)\} = af(t) + bg(t). \tag{3.31}$$

Remark 3.56 (On the linearity of the inverse Laplace transform)*.* The linearity of the inverse Laplace transform makes sure that we solve linear differential equations with the Laplace transformation method 3.3.2.. It also allows us to construct term-by-term solutions.

Theorem 3.57 (Laplace transform of the Caputo fractional derivative [14][p. 105])**.** *Let $\alpha > 0$ and $n = \lceil \alpha \rceil$. Suppose $f \in C^n[0, \infty)$ and that f satisfies the conditions for the Laplace transform. Then the Laplace transform of the Caputo fractional derivative ${}_0^C D_t^\alpha f(t)$ is given by:*

$$\mathcal{L}\left\{ {}_a^C D_0^\alpha f(t) \right\}(s) = s^\alpha \tilde{f}(s) - \sum_{k=0}^{n-1} s^{\alpha-1-k} f^{(k)}(0). \tag{3.32}$$

Theorem 3.58 (Laplace transform of the Riemann-Liouville fractional derivative [14][p. 106])**.** *Let $\alpha > 0$ and let $n = \lceil \alpha \rceil$. Suppose f is a function satisfying the conditions for the Laplace transformation and such that the fractional integrals:*

$$(I_0^k f)(0) \quad \textit{exist for } k = 0, 1, \ldots, n-1. \tag{3.33}$$

Then the Laplace transform of the Riemann–Liouville fractional derivative ${}_0^{RL} D_t^\alpha f(t)$ is given by:

$$\mathcal{L}\left\{ {}_0^{RL} D_t^\alpha f(t) \right\}(s) = s^\alpha \tilde{f}(s) - \sum_{k=0}^{n-1} s^{\alpha-1-k} (I_0^k f)(0). \tag{3.34}$$

Remark 3.59. Although both Laplace transforms of the Caputo and Riemann–Liouville derivatives share the same term $s^\alpha \tilde{f}(s)$, they differ in the form of initial conditions. The Caputo transform involves the classical derivatives $f^{(k)}(0)$, while the Riemann–Liouville transform uses fractional integrals $(I_0^k f)(0^+)$.

Theorem 3.60 (Inverse Laplace transform of the two-parameter Mittag-Leffler function [14][p.45])**.** *Let $\alpha > 0$, $\beta \in \mathbb{R}$, and $\lambda \in \mathbb{C}$. Then the inverse Laplace transform of the mittag-leffler function is,*

$$\mathcal{L}^{-1}\left\{\frac{s^{\alpha-\beta}}{s^\alpha - \lambda}\right\}(t) = t^{\beta-1}E_{\alpha,\beta}(\lambda t^\alpha),$$

Theorem 3.61 (Inverse Laplace transform of a power function [**?**, p. 23])**.** *Let $\mu > 0$, then the inverse Laplace transform of the function $s^{-\mu-1}$ is given by:*

$$\mathcal{L}^{-1}\left\{\frac{1}{s^{\mu+1}}\right\}(t) = \frac{t^\mu}{\Gamma(\mu+1)}, \quad t > 0. \tag{3.35}$$

Remark 3.62 (Application in solving fractional differential equations)*.* The Laplace transform and its inverse will play a central role in solving linear fractional differential equations in section 4.3..

3.3.3. Equilibrium/fixed points and stability

Stability analysis in fractional differential equations is extended by classical methods through the fractional derivatives. Consider the autonomous Caputo-type system:

$${}_a^C D_t^\alpha x(t) = f(x(t)), \quad x(0) = x_0,$$

where $0 < \alpha \leq 2$. An equilibrium point $x^* \in \mathbb{R}^n$ is the point when $f(x^*) = 0$.

Definition 3.63 (Stability fractional solutions [2, p. 3])**.** Let $\varphi(t, x_0)$ be the solution to the FDE. The equilibrium $x = 0$ is called stable if for every $\varepsilon > 0$, there exists $\delta > 0$ such that $\|x_0\| < \delta$ implies $\|\varphi(t, x_0)\| < \varepsilon$ for all $t \geq 0$.

It is called asymptotically stable if it is stable and $\lim_{t\to\infty} \varphi(t, x_0) = 0$.

Theorem 3.64 (Matignon's criterion for linear FDEs [2], p. 7)**.** *Consider the linear system:*

$${}_a^C D_t^\alpha x(t) = Ax(t). \qquad 0 < \alpha \leq 2.$$

The equilibrium $x = 0$ is asymptotically stable if and only if all eigenvalues $\lambda \in \sigma(A)$ satisfy $|\arg(\lambda)| > \alpha\pi/2$. Or it is stable if all eigenvalues satisfy $|\arg(\lambda)| \geq \alpha\pi/2$.

Theorem 3.65 (Stability condition for scalar FDEs [2, p. 7])**.** *The solution $x(t) \to 0$ as $t \to \infty$ if and only if $|\arg(\lambda)| > \alpha\pi/2$. Thus, the scalar Caputo equation is $O(t^{-\alpha})$ asymptotically stable under the condition as in the matrix case 3.64.*

Chapter 4

Analytical approaches to fractional differential equations

4.1. General framework

Now that we have established a basic understanding of fractional derivatives and differential equations, the next natural step is to study how such systems can be solved. Analytical methods provide a valuable insight into the behaviour of fractional differential equations (FDE's). In particular, in understanding how properties like memory and non-locality influence the system. Although closed-form solutions are rare and mostly restricted to specific cases, we can find them with the right approach. The most notably approaches are the Laplace transform, solutions involving the Mittag-Leffler and fractional power series.

First, analytical solutions give us a deeper understanding of the mathematical structure behind fractional systems. Second, they allow us to benchmark and validate the numerical methods discussed in later chapters. Finally, analytical expressions clarify the role of fractional-order parameters and are important in the study of properties such as stability and regularity.

4.2. Solutions involving the Mittag-Leffler function

One of the most fundamental tools in solving linear fractional differential equations is the Mittag-Leffler function. As we have seen in the chapter on mathematical tools, this special function is a generalization of the exponential function and naturally appears in the solutions of fractional differential equations. By the nature of the extension we get a strong connection to the Caputo and Riemann-Liouville derivatives. In this section we will discuss its role in solving linear FDE's with constant coefficients, we will show where it emerges as the eigenfunction of the different fractional differential operators, how the Mittag-Leffler function is used to construct exact solutions to FDE's, how its behavior influences the dynamics, and why it is essential in the analysis and practical modeling of the fractional harmonic oscillator.

4.2.1. *The eigenfunction property of $E_{\alpha,\beta}(\lambda t^\alpha)$ under Caputo and Riemann-Liouville derivatives*

In normal differential equations, the exponential function $e^{\lambda t}$ plays role as the eigenfunction of the derivative operator. This is a function satisfying $\frac{d}{dt}f(\lambda t) = \lambda f(\lambda t)$. In Fractional Differential Equations this role as eigenfunction is taken over by the Mittag-Leffler function. To be more specific, $E_{\alpha,1}(\lambda t^\alpha)$ behaves as an eigenfunction of the Caputo and Riemann-Liouville fractional derivatives. This property makes it a powerful tool for solving linear fractional differential equations with constant coefficients.

Remark 4.1 (Mittag-Leffler function as eigenfunction)*.* As established in theorem 3.43 of chapter 3, the Mittag-Leffler function $E_{\alpha,1}(\lambda(t-a)^\alpha)$ satisfies the following relation:

$$ {}^{C,GL}_{a}D^{\alpha}_{t}\left[E_{\alpha,1}\left(\lambda(t-a)^{\alpha}\right)\right] = \lambda E_{\alpha,1}\left(\lambda(t-a)^{\alpha}\right). \tag{4.1}$$

Showing that it is in fact an eigenfunction of the Caputo and Riemann-Liouville fractional derivative operator.

Theorem 4.2 (Caputo solution simple first-order fde)**.** *Consider the initial value problem involving the Caputo (or Riemann–Liouville) fractional derivative:*

$$ {}^{C}_{0}D^{\alpha}_{t}x(t) = \lambda x(t), \quad x(0) = x_0. \tag{4.2}$$

Where $0 < \alpha \leq 1$ and $\lambda \in \mathbb{R}$. If we want to find an analytical solution to this equation, we can use the Mittag-Leffler from remark 4.1.

$$x(t) = E_{\alpha,1}(\lambda t^\alpha).$$

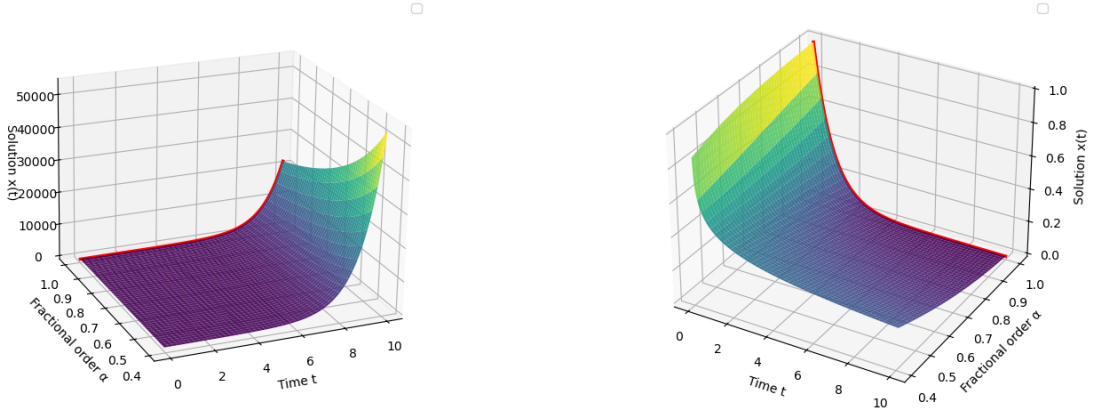
Where we pick $x(t)$ as candidate solution of the differential equation, it only remains to solve it for the intial condition. Which gives the unique solution,

$$x(t) = x_0 E_{\alpha,1}(\lambda t^\alpha).$$

Remark 4.3 (Initial conditions and coincidence of Caputo and Riemann-Liouville solutions)*.* In the example above, the initial value problem involving the Caputo or Riemann-Liouville fractional derivative have the solution expressed in terms of the Mittag-Leffler function. This example is a case in which both formulations lead to the same analytical expression, provided that the initial conditions are correctly specified for each derivate which we discussed in 3.3.. The coincidence is a direct consequence of the fact that the Mittag-Leffler function $E_{\alpha,1}(\lambda t^\alpha)$ satisfies identical fractional differential equations under both derivatives 3.44.

Remark 4.4 (Long term behaviour $E_{\alpha,1}(\lambda t^\alpha)$)*.* The solution $x(t) = E_{\alpha,1}(\lambda t^\alpha)$, as we can see in 4.1, has faster than exponential growth when $\alpha < 1$. This comes from the asymptotic expansion of the Mittag-Leffler function, as given in 3.16:

$$E_{\alpha,1}(\lambda t^\alpha) \sim \frac{1}{\alpha}\exp(\lambda^{1/\alpha} t). \tag{4.3}$$



(a) $x(t) = E_\alpha(t^\alpha)$ (b) $x(t) = E_\alpha(-t^\alpha)$

Figure 4.1: Solutions of the fractional differential equation ${}_0^C D_t^\alpha x(t) = \lambda x(t)$ for $0 < \alpha < 1$. The red curve shows the classical exponential $e^{\lambda t}$ recovered when $\alpha = 1$.

The growth is faster for smaller values of α because the term stretches the behaviour of the function more. On the other hand, when $\lambda < 0$, the Mittag-Leffler function exhibits decay, which comes from theorem 3.16:

$$E_{\alpha,1}(-\lambda t^\alpha) \sim \sum_{k=1}^{N} \frac{(-1)^{k+1}}{\lambda^k t^{\alpha k} \Gamma(1-\alpha k)} + o\left(\frac{1}{t^{\alpha(N+1)}}\right). \tag{4.4}$$

This decay is algebraic, which implies that solutions decay slowly, unlike the fast exponential decay of the classical case. This is one of the first fundamental points where classical dynamics and fractional dynamics diverge significantly.

Remark 4.5 (Difference plot and long term behaviour). Although the function $E_{\alpha,1}(-\lambda t^\alpha)$ decays more steeply than the classical exponential $e^{-\lambda t}$ for small t and small α, as we can see in figure 4.1, its long-term behaviour in the growth of the decay is different. Thus, the behaviour we get is a fast initial drop followed by a long, memory-driven tail.

4.2.2. Analytical solutions to simple higher-order FDE's

In the preceding subsection, we demonstrated how we can find in the most natural way solution to the first order fractional differential equation,

$${}_0^C D_t^\alpha x(t) = \lambda x(t), \qquad x(0) = x_0, \quad x'(0) = 0, \tag{4.5}$$

We are now going to extend this approach to the more general case $0 < \alpha < 2$. This will require us more initial data, where we first only needed $x(0) = x_0$, we now also the additional condition $x'(0) = x_1$. The following section will discuss the behaviour of simple higher-order fractional systems involving the two-parameter Mittag-Leffler function $E_{\alpha,\beta}$. We will see a more dynamic behavior, and some form of damped oscillations. This is where our analysis of the fractional oscillator truly starts taking form.

Remark 4.6 (Initial conditions and the role of the order $1 < \alpha < 2$)*.* When looking at the equation ${}^{C}D_t^{\alpha}x(t) = \lambda x(t)$, it is important to differentiate between two cases $0 < \alpha \leq 1$ and $1 < \alpha < 2$. As we have seen in section 3.3.1. on operational framework, we require different numbers of initial conditions. When $\alpha \leq 1$, we only need $x(0)$, while for $\alpha < 1$ we need both $x(0)$ and $x'(0)$. This is why the full interval $0 < \alpha < 2$ will have an extension of the first-order solutions within its fractional framework.

Theorem 4.7 (Analytical solution of the linear FDE's where $1 < \alpha \leq 2$)**.** *We consider the Caputo initial value problem:*

$$ {}_{0}^{C}D_t^{\alpha}x(t) = \lambda x(t), \quad x(0) = x_0, \quad x'(0) = x_1. $$

where $1 < \alpha \leq 2$ *and* $\lambda \in \mathbb{R}$*. Then the unique solution is given by:*

$$ x(t) = x_0 E_{\alpha,1}(\lambda t^{\alpha}) + x_1 t E_{\alpha,2}(\lambda t^{\alpha}) $$

where $E_{\alpha,\beta}(z)$ *is the two-parameter Mittag-Leffler function 3.7.*

Remark 4.8 (Role of the eigenfunction in the Solution)*.* The solutions are constructed in the same way as the the first-order case $0 < \alpha \leq 1$, using the Mittag-Leffler function $E_{\alpha,1}(\lambda t^{\alpha})$,

$$ {}_{0}^{C}D_t^{\alpha}\left[E_{\alpha,1}(\lambda t^{\alpha})\right] = \lambda E_{\alpha,1}(\lambda t^{\alpha}). $$

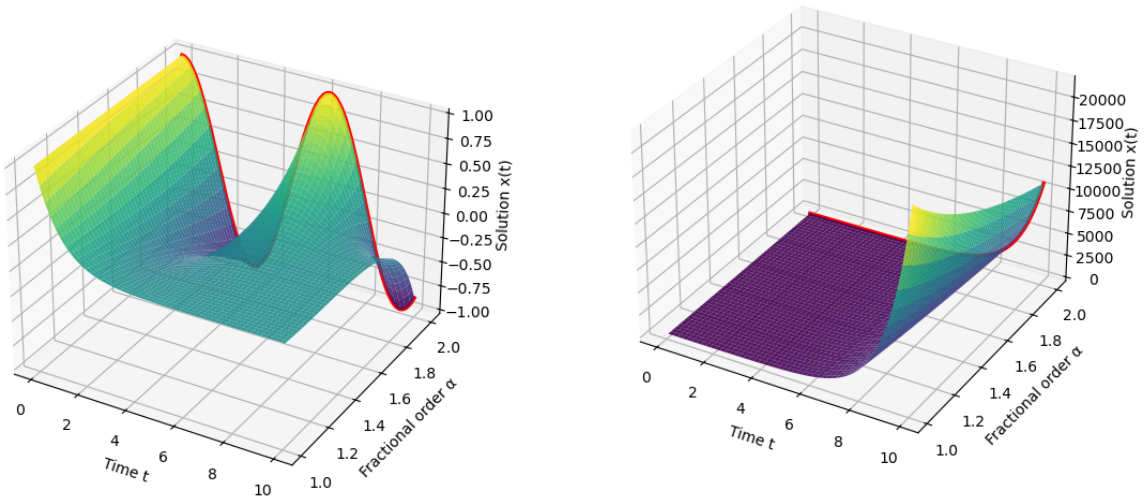
Only this time for $\alpha > 1$, we had to take care of the initial condition $x'(0) = x_1$. In classical theory, we try to construct an independent solution with the same "root"; we normally do this by multiplying with t taking into account the derivative. In fractional setting, we can look at the derivative of the Mittag-Leffler function. This makes it that the solution remains within the Mittag-Leffler function family, only modified to some power of t dependent of the order of α.

Corollary 4.9 (Extension framework, solution for $0 < \alpha < 2$)**.** *Let* $0 < \alpha < 2$*,* $\lambda \in \mathbb{R}$*, and consider the Caputo initial value problem:*

$$ {}_{0}^{C}D_t^{\alpha}x(t) = \lambda x(t), \quad x(0) = x_0, \quad x'(0) = x_1. $$

Then the unique solution is:

$$ x(t) = \begin{cases} x_0\, E_{\alpha,1}(\lambda t^{\alpha}), & 0 < \alpha \leq 1, \\ x_0\, E_{\alpha,1}(\lambda t^{\alpha}) + x_1 t\, E_{\alpha,2}(\lambda t^{\alpha}), & 1 < \alpha < 2. \end{cases} $$



(a) $\lambda = -1$ (b) $\lambda = 1$

Figure 4.2: Solutions of the fractional differential equation ${}_0^C D_t^\alpha x(t) = \lambda x(t)$ for $1 < \alpha \leq 2$. The behavior reflects oscillatory decay for negative λ and Mittag-Leffler-type growth for positive λ.

Remark 4.10 (Asymptotic growth for higher-order fractional equation $\lambda > 0$). If we look at the solution of the higher-order fractional differential equation for $\lambda > 0$:

$$x(t) = x_0\, E_{\alpha,1}(\lambda t^\alpha) + x_1\, t\, E_{\alpha,2}(\lambda t^\alpha).$$

We can see that there is a rapid growth in figure **??**. The asymptotic behaviour of the Mittag-Leffler function, on the other hand is a little different for positive λ:

$$E_{\alpha,1}(\lambda t^\alpha) \sim \frac{1}{\alpha} \exp(\lambda^{1/\alpha} t), \qquad t\, E_{\alpha,2}(\lambda t^\alpha) \sim \frac{1}{\alpha} \lambda^{-1/\alpha} \exp(\lambda^{1/\alpha} t),$$

as seen in theorem 3.16. This makes the solution behave in the following way:

$$x(t) \sim \frac{1}{\alpha} \left(x_0 + x_1 \lambda^{-1/\alpha}\right) \exp(\lambda^{1/\alpha} t), \quad t \to \infty.$$

Which tells us that its grows exponentially dependent on the fractional order α.

Remark 4.11 (Algebraic decay for higher-order fractional equation $\lambda < 0$). On the other hand, when $\lambda < 0$, we get that the solution:

$$x(t) = x_0\, E_{\alpha,1}(-\lambda t^\alpha) + x_1\, t\, E_{\alpha,2}(-\lambda t^\alpha),$$

decays algebraically rather than exponentially. In the same sense as in the first-order case when ($0 < \alpha < 1$). Using the asymptotic expansion for negative values gives us/;

$$E_{\alpha,\beta}(-\lambda t^\alpha) \sim \sum_{k=1}^{N} \frac{(-1)^{k+1}}{\lambda^k t^{\alpha k} \Gamma(\beta - \alpha k)}, \quad t \to \infty.$$

So the term decays roughly like $\mathcal{O}(t^{-\alpha})$.The decay is slower and reflects the memory effect of the fractional dynamics.

Remark 4.12 (Origin of oscillations in fractional systems)*.* Oscillations in fractional systems arise not from classical sine and cosine terms, but from the alternating sign in the Mittag-Leffler series for the negative arguments. Combining this with the algebraic decay and memory effects of the fractional derivatives, produces slowly damped, non-trigonometric oscillations. The parameter α controls here both the rate of decay and the phase delay, leading to a distinction in behaviour compared to the classical second-order differential equation 2.1.

4.3. Laplace transform methods

While the use of Mittag-Leffler functions in previous sections may look like educated guessing or function fitting so to say, these solutions are in fact connected to another framework, the Laplace transformations. The Laplace transform method gives a systematic and thorough approach to solving fractional differential equations by transforming them into algebraic equations. It also helps us confirm the validity of Mittag-Leffler based solutions.

4.3.1. Solving linear FDE's via the Laplace transform

In the previous solutions using Mittag-Leffler functions we made well informed guesses using the eigenfunctions. In the Laplace transform method we will show how solutions arise from the analytic structure of the fractional differential equations. For the linear differential equations,

$$D^\alpha x(t) = \lambda x(t), \quad x(0) = x_0, \quad 0 < \alpha \leq 1, \quad \lambda \in \mathbb{R},$$

where D^α denotes either the Caputo or Riemann-Liouville fractional derivative, the Laplace transform offers a systematic method for finding exact solutions.

Remark 4.13 (General solution method)*.* In general, fractional differential equations are solved via the Laplace transform by first applying the transform to the system. This transforms the fractional derivatives into an algebraic expressions involving the unknown function and its initial values. We then isolate the transformed expressionm, such that we can apply the inverse Laplace transform to obtain the solution.

Remark 4.14 (Solutions and Mittag-Leffler function)*.* In many cases, where the system has constant coefficients, the solutions naturally involves terms of the Mittag-Leffler function. As we have seen in section 4.2..

Example 4.15 (Method for Caputo Simple fractional oscillator $0 < \alpha < 1$)**.** In our first demonstration of this method, we look at the simple FDE:

$$ {}_0^C D_t^\alpha x(t) = \lambda x(t), \quad x(0) = x_0, \quad 0 < \alpha \leq 1, \quad \lambda \in \mathbb{R},$$

First, we apply the Laplace transformation to both sides using the formula for the Caputo derivative 3.57:

$$\mathcal{L}\left\{{}^C D^\alpha x(t)\right\}(s) = s^\alpha \tilde{x}(s) - s^{\alpha-1} x_0,$$

This gives the algebraic equation:

$$s^\alpha \tilde{x}(s) - s^{\alpha-1} x_0 = \lambda \tilde{x}(s).$$

Solving for $\tilde{x}(s)$, we obtain:

$$\tilde{x}(s) = \frac{s^{\alpha-1} x_0}{s^\alpha - \lambda}.$$

Taking the inverse Laplace transform 3.60, we use:

$$\mathcal{L}^{-1}\left\{\frac{s^{\alpha-1}}{s^\alpha-\lambda}\right\}(t) = E_{\alpha,1}(\lambda t^\alpha), \quad \beta = 1.$$

Thus, the final solution is:

$$x(t) = x_0 E_{\alpha,1}(\lambda t^\alpha).$$

Remark 4.16 (Coincide of the Caputo and Riemann-Liouville Laplace transformation). The same solution holds if D^α is the Riemann-Liouville derivative, provided the initial conditions are adapted to the fractional integral I 3.45.

Example 4.17 (Method for Caputo simple fractional oscillator $1 < \alpha < 2$)**.** We now consider the same equation, but with a higher fractional order:

$${}_0^C D_t^\alpha x(t) = \lambda x(t), \quad x(0) = x_0, \quad x'(0) = x_1, \quad 1 < \alpha < 2.$$

Applying the Laplace transform for this order gives:

$$\mathcal{L}\left\{{}_0^C D^\alpha x(t)\right\}(s) = s^\alpha \tilde{x}(s) - s^{\alpha-1}x_0 - s^{\alpha-2}x_1.$$

Where we recognize an extra term dependent on the second initial condition that the higher order α provides us. In the system this looks like,

$$s^\alpha \tilde{x}(s) - s^{\alpha-1}x_0 - s^{\alpha-2}x_1 = \lambda \tilde{x}(s).$$

Where we recognize an extra term due to the second initial condition required for $\alpha > 1$. The Laplace-transformed equation becomes:

$$\tilde{x}(s) = \frac{s^{\alpha-1}x_0 + s^{\alpha-2}x_1}{s^\alpha - \lambda}.$$

We can now apply the inverse Laplace transform 3.60 term by term:

$$x(t) = x_0 E_{\alpha,1}(\lambda t^\alpha) + x_1 t E_{\alpha,2}(\lambda t^\alpha).$$

Remark 4.18 (Extension to higher-order Fractional simple oscillators). For values of α greater than 2, the solution method proceeds in a similar fashion. By including n initial conditions in the Caputo sense, $x^{(k)}(0) = x_k$ for $k = 0, 1, \ldots, n-1$, we obtain a general solution of the form:

$$x(t) = \sum_{k=0}^{n-1} x_k\, t^k E_{\alpha,k+1}(\lambda t^\alpha),$$

where each term consists of a power of t multiplied by a Mittag-Leffler function.

Remark 4.19 (Connection with two-term fractional equations). Now that we have a general procedure for obtaining exact and closed-form solutions to one-term fractional differential equations, it is useful to extend the structure to two-term fractional equations. These take the form:

$${}_0^C D_t^\alpha x(t) + b\ {}_0^C D_t^\beta x(t) = 0, \quad 1 < \beta < \alpha \leq 2.$$

The Laplace transform method remains usable in this system and gives solutions expressed in terms of the two-parameter Mittag-Leffler functions $E_{\alpha,\beta}$.

Theorem 4.20 (Solution two-term homogeneous FDE [**?**, p. 50])**.** *Let $1 < \alpha \leq 2$, $0 < \beta < 1$, and $b \in \mathbb{R}$. Consider the Caputo-type two-term fractional differential equation:*

$$ {}_0^C D_t^\alpha x(t) + b\, {}_0^C D_t^\beta x(t) = 0, \quad x(0) = x_0, \quad x'(0) = x_1. $$

Then the unique solution is given by:

$$ x(t) = x_0 E_{\gamma,1}(-bt^\gamma) + x_1\, t E_{\gamma,2}(-bt^\gamma), $$

where $\gamma = \alpha - \beta$.

Example 4.21 (Solution of a two-term fractional differential equation via Laplace transform)**.** Consider the initial value problem:

$$ {}_0^C D^\alpha x(t) + b\, {}_0^C D^\beta x(t) = 0, \quad x(0) = x_0, \quad x'(0) = x_1, $$

with $1 < \alpha \leq 2$, $0 < \beta < 1$, and $b \in \mathbb{R}$. By applying the Laplace transform to both sides gives:

$$ s^\alpha \tilde{x}(s) - s^{\alpha-1} x_0 - s^{\alpha-2} x_1 + b\left(s^\beta \tilde{x}(s) - s^{\beta-1} x_0\right) = 0. $$

Isolating $\tilde{x}(s)$:

$$ \tilde{x}(s) = \frac{x_0(s^{\alpha-1} + bs^{\beta-1}) + x_1 s^{\alpha-2}}{s^\alpha + bs^\beta}. $$

Let $\gamma = \alpha - \beta$, we can manipulate the expression into:

$$ \tilde{x}(s) = x_0 \left(\frac{s^{\gamma-1}}{s^\gamma + b} + \frac{bs^{-1}}{s^\gamma + b} \right) + x_1 \left(\frac{s^{\gamma-2}}{s^\gamma + b} \right). $$

Applying the inverse Laplace transform term by term gives:

$$ x(t) = x_0 E_{\gamma,1}(-bt^\gamma) + x_1 t E_{\gamma,2}(-bt^\gamma). $$

Remark 4.22 (On the discarding of singular terms)*.* When we are formulating the differential equation in the Laplace-domain, we notice that we have a term $\frac{s^{-1}}{s^\gamma}$, which becomes problematic because of the function divergence at $\Gamma(0)$. This corresponds to a non-integrable singularity at $t = 0$ in fig. 3.1. Therefore, we discard this term in the solution since it is not valid in the sense of the existence of the fractional derivative.

4.3.2. Solving nonlinear FDE's via the Laplace transform

Remark 4.23 (Limitations for nonlinear FDE's)*.* As discussed in the previous remark, is the Laplace transformation an effective framework for solving linear fractional differential equations (FDEs). However, the application to nonlinear problems is more constrained. Nonlinear terms do not have straightforward transformation into algebraic expressions, making the method less directly applicable.

Example 4.24 (Transforming a nonlinear FDE to an integral equation)**.** If we look at a nonlinear Caputo FDE of order $0 < \alpha < 1$:

$$ {}_0^C D^\alpha x(t) = -x(t)^3, \quad x(0) = x_0. $$

Applying the Laplace transform gives:

$$ s^\alpha \tilde{x}(s) - s^{\alpha-1} x_0 = -\mathcal{L}\{x(t)^3\}(s). $$

This equation cannot be solved directly for $\tilde{x}(s)$ since we do not know the nature of x. However, we can apply the inverse Laplace transformation on:

$$ \tilde{x}(s) = \frac{s^{\alpha-1} x_0 - \mathcal{L}\{x(t)^3\}(s)}{s^\alpha}. $$

Recognizing the inverse power function 3.61 to get the simpler form:

$$ x(t) = x_0 - \mathcal{L}^{-1}\left\{ \frac{1}{s^\alpha} \mathcal{L}\{x(t)^3\}(s) \right\}. $$

Remark 4.25 (Analytical and numerical implications of nonlinear FDEs)*.* If we have non-linear fractional differential equations such as the example 4.24, it can ben challenging to compare it with its linear counterpart. In analytical terms we cannot use the Mittag-Leffler function or any other well-known function. Therefore, when looking at the behavior and stability we need other techniques, one of which is numerical approximation.

4.4. Fractional power series methods

4.4.1. Fractional power series method

The Fractional Power Series (FPS) extends the classical series method for solving differential equations to the fractional world. The method uses non-integer powers provided from the order of the fractional derivative to match the powers of the series.

Definition 4.26 (Fractional power series)**.** Let $\gamma > 0$ and $\{c_k\}_{k=0}^\infty \subset \mathbb{R}$ be a sequence of coefficients. The fractional power series (FPS) centered at $t = 0$ is a series of the form:

$$ x(t) = \sum_{k=0}^{\infty} c_k t^{\gamma k}. $$

Remark 4.27 (On convergence of fractional power series)*.* The convergence of a fractional power series depends on the growth of its coefficients c_k. However, in nonlinear or rapidly growing recursions, the convergence may be limited to a finite interval.

Lemma 4.28 (Radius of convergence for fractional power series)**.** *Let* $x(t) = \sum_{k=0}^{\infty} c_k t^{\gamma k}$ *be a fractional power series with* $\gamma > 0$ *and coefficients* $\{c_k\} \subset \mathbb{R}$*. Then the radius of convergence* $R > 0$ *is given by:*

$$\frac{1}{R_z} = \limsup_{k\to\infty} |c_k|^{1/k}.$$

Proof. We define the substitution $z = t^{\gamma}$, under which the fractional power series becomes a classical power series:

$$x(t) = \sum_{k=0}^{\infty} c_k z^k.$$

By the classical Cauchy–Hadamard formula [**?**, Sec. 8], the radius of convergence R_z of this power series in z is:

$$\frac{1}{R_z} = \limsup_{k\to\infty} |c_k|^{1/k}.$$

In the fractional scene we have $z = t^{\gamma}$, thus the series converges for all $|t| < R$, where $R = R_z^{1/\gamma}$. Therefore our convergence radius becomes:

$$\frac{1}{R} = \left(\limsup_{k\to\infty} |c_k|^{\frac{1}{\gamma k}} \right).$$

□

Remark 4.29 (Classical against fractional power series)*.* Here, the classical power series is an expansion based on the derivative at a point and uses integer powers. Is the fractional power series quite different, the expansion is based on non-integer powers„ and the expansions is not based on the derivative in a point like the Taylor series. The fractional series is a custom construct that depends on the structure of the underlying fractional differential equation.

Theorem 4.30 (Caputo derivative of a fractional power series)**.** *Let* $f(t) = \sum_{k=0}^{\infty} c_k t^{\gamma k}$ *be the fractional power series with* $\gamma > 0$*, and let* $\alpha > 0$*. Suppose the series converges for* $t \in [0, T)$*. Then the Caputo fractional derivative of* f *is given by:*

$${}_0^C D_t^{\alpha} f(t) = \sum_{k=n}^{\infty} c_k \frac{\Gamma(\gamma k + 1)}{\Gamma(\gamma k + 1 - \alpha)} t^{\gamma k - \alpha},$$

where $n = \lceil \alpha \rceil$*. The starting point* k *depends on the existence of the derivative of the power series which we discussed in theorem 3.41 (where the derivative of the Caputo and Riemann-Liouville derivative coincide).*

Remark 4.31 (Framework for solving FDEs with FPS)*.* To solve a fractional differential equation using the fractional power series (FPS) method, we assume that a solution of the form of a power series exists and is sufficiently smooth. We then substitute this series into the differential equation. Next, we correct for the translated index of the summation and pick a γ that suits our structure.

In the case of a simple homogeneous differential equation with one order, it is convenient to choose a γ equal to the fractional order. This makes it simpler to align powers and find a recursive relationship. Once we found the coefficients of the recursive relation we have the iterative solution to our system.

Example 4.32 (Power series simple fractional oscillator)**.** Lets consider the standard fractional oscillator:

$$ {}_0^C D_t^\alpha x(t) = \lambda x(t), \quad x(0) = c_0, $$

with $0 < \alpha \leq 1$. Assuming the FPS solution is of the form $x(t) = \sum_{k=0}^{\infty} c_k t^{k\gamma}$ and substituting into the equation:

$$ \sum_{k=1}^{\infty} c_k \frac{\Gamma(\gamma k + 1)}{\Gamma(\gamma k + 1 - \alpha)} t^{\gamma k - \alpha} = \sum_{k=0}^{\infty} c_k t^{k\gamma}. $$

We coveniently pick our γ to align our powers for simplification of the structrue. In this case $\gamma = \alpha$:

$$ \sum_{k=1}^{\infty} c_k \frac{\Gamma(\alpha k + 1)}{\Gamma(\alpha k + 1 - \alpha)} t^{\alpha(k-1)} = \sum_{k=0}^{\infty} c_k t^{k\alpha}. $$

Correcting for the shifted index, we equate coefficients and find the recursive relation:

$$ c_{k+1} = -\lambda c_k \cdot \frac{\Gamma(k\alpha + 1)}{\Gamma((k+1)\alpha + 1)}. $$

Applying this recurrence in the series gives the solution:

$$ x(t) = c_0 \sum_{k=0}^{\infty} \frac{(-\lambda t^\alpha)^k}{\Gamma(k\alpha + 1)} = c_0 E_{\alpha,1}(-\lambda t^\alpha), $$

Remark 4.33 (Comparison framework)*.* The FPS method provides a constructive procedure for solving fractional differential equations, particularly when the Laplace or Mittag-Leffler methods are not usable due to nonlinearity or irregular terms. The downside of this method is the convergence that we must keep in mind and the alignment of the order of the power functions can be problematic.

4.4.1.1. Higher order homogeneous fractional differential equation

Definition 4.34 (Higher order fractional oscillator)**.** The higher order fractional differential equation is given by:

$$ {}_0^C D_t^\alpha x(t) + \nu \, {}_0^C D_t^\beta x(t) + \delta x(t) = 0, $$

where $0 < \beta < 1$ and $1 < \alpha \leq 2$ are the orders of the fractional derivatives, and $\nu, \delta \in \mathbb{R}$ the constants terms.

Example 4.35 (Solving a three-term fde using the fractional power series method)**.** Consider the higher-order fractional differential equation from Definition 4.34:

$$ {}_0^C D_t^{\alpha} x(t) + \nu \, {}_0^C D_t^{\beta} x(t) + \delta x(t) = 0, \quad x(0) = c_0, \quad x'(0) = c_1. $$

We can use the power series to find a solution to this problem. In line with the framework that we gave in remark 4.31 we assume that the solution has the form:

$$ x(t) = \sum_{k=0}^{\infty} c_k t^{\gamma k}, $$

where $\gamma > 0$ will be chosen appropriately later on. First, we use theorem 4.30, we obtain the terms for the fractional derivative:

$$ {}_0^C D_t^{\mu} x(t) = \sum_{k=k_\mu}^{\infty} c_k \frac{\Gamma(\gamma k + 1)}{\Gamma(\gamma k + 1 - \mu)} t^{\gamma k - \mu}, \quad \text{for } \mu \in \{\alpha, \beta\}, $$

where $k_\mu = \{1, 2\}$ dependent on the order. Second, we substitute the derivatives into the differential equation:

$$ \sum_{k=2}^{\infty} c_k \frac{\Gamma(\gamma k + 1)}{\Gamma(\gamma k + 1 - \alpha)} t^{\gamma k - \gamma} + \nu \sum_{k=1}^{\infty} c_k \frac{\Gamma(\gamma k + 1)}{\Gamma(\gamma k + 1 - \beta)} t^{\gamma k - \beta} + \delta \sum_{k=0}^{\infty} c_k t^{\gamma k} = 0. $$

Third, the re-indexation of the summations and distribution of the powers:

$$ t^{2\gamma - \alpha} \sum_{k=0}^{\infty} c_{k+2} \frac{\Gamma(\gamma(k+2)+1)}{\Gamma(\gamma(k+2)+1-\alpha)} t^{\gamma k} + \nu t^{\gamma - \beta} \sum_{k=0}^{\infty} c_{k+1} \frac{\Gamma(\gamma(k+1)+1)}{\Gamma(\gamma(k+1)+1-\beta)} t^{\gamma k} + \delta \sum_{k=0}^{\infty} c_k t^{\gamma k} = 0. $$

Fourth, finding the recursive relation:

$$ c_k = -\frac{1}{\delta} \left[t^{2\gamma - \alpha} c_{k+2} \frac{\Gamma(\gamma(k+2)+1)}{\Gamma(\gamma(k+2)+1-\alpha)} + \nu \, t^{\gamma - \beta} c_{k+1} \frac{\Gamma(\gamma(k+1)+1)}{\Gamma(\gamma(k+1)+1-\beta)} \right]. $$

Lastly, we want to pick γ in such a fashion that we lose the dependence on the factor t in our recursive equation. This is when $2\gamma - \alpha = \gamma - \beta = 0$ our when $\gamma = \alpha - \beta$ such that the powers become constant factors. This gives us the solution:

$$ c_k = -\frac{1}{\delta} \left[c_{k+2} \frac{\Gamma((\alpha-\beta)(k+2)+1)}{\Gamma((\alpha-\beta)(k+2)+1-\alpha)} + c_{k+1} \frac{\nu \, \Gamma((\alpha-\beta)(k+1)+1)}{\Gamma((\alpha-\beta)(k+1)+1-\beta)} \right]. $$

Remark 4.36 (From recursion to global behaviour)*.* The recursive relationship we have found in example 4.35 shows how the the coefficients c_k form a solution to the system, but it does not immediately reveal what the global behavior of the solution $x(t)$ is. To understand whether the system exhibits decay or growth, or if there is a form of oscillation must be analyzed further.

4.5. Summary of analytical techniques

In this chapter, we introduced three analytical techniques to solve fractional differential equations; the solutions via the Mittag-Leffler function, Laplace transform methods, and the fractional power series method.

Each technique serves a different view and purpose. The Mittag-Leffler approach uses the eigenfunction structure of fractional derivatives, particularly effective for linear FDEs with constant coefficients. The Laplace transform method gives a systematic algebraic framework, enabling us to find explicit solutions. The FPS method constructs solutions through generalized series expansions, offering us flexibility in non-linear or unsolvable by other cases solutions.

In summary, these methods form a toolkit for understanding and solving FDEs, where they serve as theoretical tools and benchmarks for validating numerical approximation.

Chapter 5

Numerical approximation to fractional differential equations

In the previous chapter, we looked at analytical methods for solving fractional differential equations, where we gave a framework for the solutions expressed in terms of the Mittag-Leffler function, Laplace transforms, and fractional power series. While these methods offer insights into the structure and behaviour of fractional systems, they are limited to linear or special cases.

In practice, if we want to solve FDEs with and without linearity or homogeneity, we use numerical methods to approximate their solutions. In this chapter we will introduce two numerical techniques specifically designed for fractional derivatives, which will help us find solutions to our system. These methods significantly differ from the classical methods. We discuss the following points, the Fractional Power Series Method, which is a numerical extension of the analytical FPS approach developed in the previous chapter. Followed by the L1 Scheme, a finite difference method specifically for approximating Caputo derivatives on a uniform grid.

We also compare their performance in terms of convergence, stability, and computational cost, in order to provide guidance on which method is appropriate for different settings

5.1. Fractional power series

In Section 4.4, we derive solutions to fractional differential equations using the fractional power series method. This approach generalizes the classical series method. This method is particularly effective near the initial point $t = 0$. The infinite series representation of the solutions gives insight into the local behaviour of the solutions.

However, from a numerical standpoint, working with an infinite series is not feasible. In practice, we must truncate the expansion after a finite number of terms. This introduces an approximation error, which depends on the order of truncation and the properties of the function being approximated.

In this section, we focus on the numerical consequences of truncating the fractional power series as a numerical approximation. We analyse the form and order of the truncation error and its asymptotic behaviour. This analysis allows us to assess the usability and limitations of the method when used for numerical com-

putation.

Definition 5.1 (Truncated fractional power series)**.** Let $\gamma > 0$ and $\{c_k\}_{k=0}^{\infty} \subset \mathbb{R}$ be a sequence of coefficients. The truncated fractional power series of order N is given by:

$$x_N(t) = \sum_{k=0}^{N} c_k t^{\gamma k},$$

which approximates the full fractional power series,

$$x(t) = \sum_{k=0}^{\infty} c_k t^{\gamma k}$$

as defined in definition 4.26.

Theorem 5.2 (Truncation error power series [14, p. 169])**.** *Let $x(t)$ be the exact solution and $x_N(t)$ the truncated series of order N. Then the truncation error satisfies:*

$$|x(t) - x_N(t)| \leq O(t^{(N+1)\alpha}) \quad \textit{as } t \to 0.$$

Remark 5.3 (Local validity and accuracy)*.* The error bound shows that the method is accurate near the origin and for small values of t, but becomes less effective for large values t, which motivates the use of a minimal study interval.

Example 5.4 (Error estimate for Mittag-Leffler expansion)**.** The series expansion of the power series for the simple fractional oscillator 4.32:

$$E_\alpha(\lambda t^\alpha) = \sum_{k=0}^{N} \frac{(\lambda t^\alpha)^k}{\Gamma(\alpha k + 1)} + R_N(t).$$

Has the remainder:

$$|R_N(t)| \leq \frac{|\lambda t^\alpha|^{N+1}}{\Gamma(\alpha(N+1)+1)},$$

which confirms the $O(t^{(N+1)\alpha})$ truncation error.

Remark 5.5 (Limitations and connection to L1/L2 schemes)*.* While the truncated fractional power series gives local accurate near $t = 0$, its convergence radius is often limited due to the growth of the coefficients and the factorial nature of the denominators (see section. 4.4.).

To address this limitation we look for other methods, the L1 and L2 methods introduced in upcoming section. These methods are constructed on uniform grids and are specifically designed to handle more global behavior. This numerical method complements the power series approach by offering stability and accuracy where the FPS method becomes insufficient.

5.2. L1 and L2 method for Caputo derivatives

In the analytical framework of fractional differential equations, stands the Caputo derivative a central role due to its connection with classical initial conditions. However, its nonlocal definition gives challenges for finding exact solutions. Therefore we revert to numerical computation, when such solutions are not possible to find.

This section introduces two numerical methods used to approximate the Caputo derivative. The L1 method and the L2 method. Where the L1 method is suitable for derivatives of order $0 < \alpha < 1$ and the L2 method is designed for derivatives with $1 < \alpha < 2$.

Both methods rely on uniform time grids and a weight structure that mimic the properties that we have seen in fractional calculus.

5.2.1. L1 and L2 approximations of Caputo derivatives

Definition 5.6 (Uniform Grid for L1-Type Schemes)**.** Let the interval $[0, T]$ be split into N equal subintervals with step size $\Delta t = T/N$. We define a uniform grid of points by:

$$t_n = n\Delta t, \quad n = 0, 1, \ldots, N.$$

The L1 and L2 methods for the Caputo fractional derivative at $t = t_n$ both use this uniform grid to approximate a solution.

Remark 5.7 (Role of the uniform grid in fractional schemes)*.* A uniform grid is used to ensure that the discretized derivative does not introduce bias or irregularity due to inconsistent spacing. Uniformity guarantees stable weighting in the non-local memory terms inherent to fractional derivatives.

Definition 5.8 (L1 method for Caputo derivatives with $0 < \alpha < 1$ [10, p. 49])**.** Let $f(t)$ be Caputo fractional integrable and consider a uniform grid t_n for $n = 0, 1, \ldots, N$ with step size Δt. The Caputo derivative of order $0 < \alpha < 1$ at $t = t_n$ is approximated by:

$$\left[{}_0^C D_t^\alpha f(t)\right]_{t=t_n} \approx \sum_{k=0}^{n-1} b_{n-k-1} \left(f(t_{k+1}) - f(t_k)\right) + \mathcal{O}(\Delta t^{2-\alpha}),$$

where the weights b_k are given by:

$$b_k = \frac{\Delta t^{-\alpha}}{\Gamma(2-\alpha)} \left[(k+1)^{1-\alpha} - k^{1-\alpha}\right].$$

Remark 5.9 (Interpretation and practical use of the L1 method)*.* The L1 method is a straightforward finite difference approximation of the Caputo derivative, base on the weighted backward difference method.

Theorem 5.10 (Error estimate for the L1 method [10, p. 43-46])**.** *Let f be Caputo integratable and bounded function. For an order $0 < \alpha < 1$, the error of the L1 approximation at $t = t_n$ satisfies:*

$$\left| {}_0^C D_t^\alpha f(t_n) - \sum_{k=0}^{n-1} b_{n-k-1}(f(t_{k+1}) - f(t_k)) \right| \leq C\Delta t^{2-\alpha},$$

where is some constant $C > 0$ independent of Δt and n, but it may be dependent on T, α, and f.

Remark 5.11 (Role and limitations of the L1 method)*.* The L1 method offers a scheme for approximating Caputo derivatives of order $0 < \alpha < 1$, making it particularly effective for modeling simple one-term differential equations. Its construction relies on first-order differences, which makes it simple to implement. However, it is limited by the order of the derivative and the behaviour around the point $t = 0$. By the first limitation, we are motivated to search for higher-order schemes such as the L2 method, which we will discuss next.

Definition 5.12 (L2 method for Caputo derivatives with $1 < \alpha < 2$ [10, p.49])**.** Let $f(t)$ be Caputo fractional integrable, if we want to approximate a higher order of the Caputo derivative, order $1 < \alpha < 2$. On the same uniform grid $t_n = n\Delta t$ as before, then we use the L2 method approximation:

$$\left[{}_0^C D_t^\alpha f(t) \right]_{t=t_n} \approx \sum_{k=-1}^{n} W_k f(t_{n-k}) + \mathcal{O}(\Delta t^{3-\alpha}),$$

where the weights W_k are given by:

$$W_k = \frac{\Delta t^{-\alpha}}{\Gamma(3-\alpha)} \begin{cases} 1, & k = -1, \\ 2^{2-\alpha} - 3, & k = 0, \\ (k+2)^{2-\alpha} - 3(k+1)^{2-\alpha} + 3k^{2-\alpha} - (k-1)^{2-\alpha}, & 1 \leq k \leq n-2, \\ -2n^{2-\alpha} + 3(n-1)^{2-\alpha} - (n-2)^{2-\alpha}, & k = n-1, \\ n^{2-\alpha} - (n-1)^{2-\alpha}, & k = n. \end{cases}$$

Here are the weights W_k the product of a fixed factor $\frac{\Delta t^{-\alpha}}{\Gamma(3-\alpha)}$ multiplied by the expression that depends on k.

Theorem 5.13 (Error estimate for the L2 method [10, p. 43-46])**.** *Let f be Caputo integratable and bounded function. When looking at the L2 method for order $0 < \alpha < 1$, the error bound is given by the estimate:*

$$\left| {}_0^C D_t^\alpha f(t_n) - \sum_{k=-1}^{n} W_k f(t_{n-k}) \right| \leq C\Delta t^{3-\alpha},$$

where is some constant $C > 0$ independent of Δt and n, but it may be dependent on T, α, and f.

Remark 5.14 (Computational complexity of the L1 and L2 methods). Both the L1 and L2 methods have a computational complexity of $\mathcal{O}(N^2)$, this comes from the fact that each time step uses the summation of the previous steps. Additionally, both methods require the computation of the Gamma functions in the weights, which can add significant complexity for the computation. In practice, we precompute these weights and store them so we can apply them later on in the process.

Example 5.15 (L2 approximation of $D^\alpha x(t)$ for $x(t) = \sin(t)$)**.** If we want to calculate the fractional derivative of $x(t) = \sin(t)$ of order $1 < \alpha < 2$. We can look at the numerically approximate of $D^\alpha x(t)$. We consider the uniform grid $t_n = n\Delta t$ for $n = 0, 1, \ldots, N$. The L2 approximation at $t = t_n$ is given by:

$$\left[{}_0^C D_r^\alpha x(t)\right]_{t=t_n} \approx \sum_{k=-1}^{n} W_k \sin(t_{n-k}) + \mathcal{O}(\Delta t^{3-\alpha}),$$

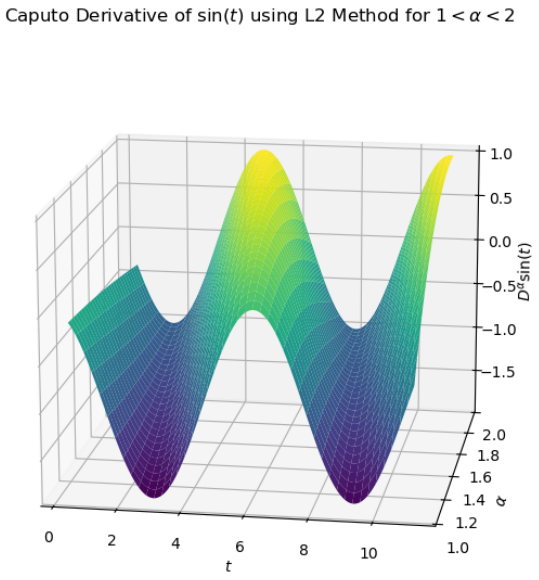


Figure 5.1: 3D plot of the L2 approximation of $D^\alpha \sin(t)$ for $1 < \alpha < 2$.

Remark 5.16 (Phase shift and damping in fractional sine derivatives). The fractional derivative ${}^C D^\alpha \sin(t)$ for $1 < \alpha < 2$ exhibits both amplitude and phase shifting compared to the classical second-order derivative $\frac{d^2}{dt^2}\sin(t) = -\sin(t)$. As α decreases from 2 to 1, the peak of the response slowly lags behind the original sine wave, which we can see as the memory effect.

Remark 5.17 (From discretization to application). The L2 method defined above are discrete analogue for the Caputo derivative. This building blocks form the foundation for numerically framework for solving one-term fractional differential equations of the form:

$${}^C D^\alpha x(t) + bx(t) = f(t).$$

In the following section, we apply these methods to construct a numerical scheme for one-term FDEs to find numerical approximations.

5.2.2. Numerical solution of one-term fractional differential equations

In this subsection, we focus on numerically solving single-term fractional differential equations of the form:

$$ {}_0^C D_t^\alpha x(t) + bx(t) = f(t), \quad x(0) = x_0, \quad x'(0) = x_1, \quad \text{for } 1 < \alpha < 2. \tag{5.1}$$

We will be doing this by using the discrete analogue of the Caputo fractional derivative in terms of the L2 method that we have seen in the last subsection.

Theorem 5.18 (L2 scheme for one-term inhomogeneous fractional differential equation)**.** *Consider the fractional differential equation:*

$$ {}_0^C D_t^\alpha x(t) + bx(t) = f(t), \quad x(0) = x_0, \quad x'(0) = x_1, $$

with $1 < \alpha < 2$ and $b \in \mathbb{R}$. Let $t_n = n\Delta t$, where $\Delta t = T/N$ is a uniform time step. Then the L2 numerical scheme to approximating the solution $x(t_n) \approx x_n$ is given by:

$$ \sum_{k=-1}^{n} W_k x_{n-k} + bx_n = f(t_n). $$

If we rewrite this to make an iterative relation with leading term x_{n+1}:

$$ x_{n+1} = \frac{1}{W_{-1}} \left(f(t_n) - \sum_{k=0}^{n} W_k \, x_{n-k} - bx_n \right). $$

Remark 5.19 (From discretization to application)*.* In the same fashion as our one-term differential equation can we look at the L1 and L2 methods are discrete analogues for Caputo derivatives of orders $0 < \beta < 1$ and $1 < \alpha < 2$. These methods form the core for numerically solving multi-term fractional differential equations. In the following section, we apply these methods to construct the complexer numerical scheme for two-term FDEs of the form:

$$ {}^C D^\alpha x(t) + a \, {}^C D^\beta x(t) + bx(t) = f(t). $$

5.2.3. Numerical solution of two-term fractional differential equations

In the previous section, we introduced the L1 and L2 methods as numerical approximations for the Caputo fractional derivative on a uniform grid. We are now going to apply these methods to the numerical solution of non-homogeneous two-term fractional differential equations of the form:

$$ {}^C D^\alpha x(t) + a \, {}^C D^\beta x(t) + bx(t) = f(t), \tag{5.2}$$

where $1 < \alpha < 2$, $0 < \beta < 1$, and $a, b \in \mathbb{R}$. A key difference between what we have seen before and what we do now is the introduction of the function $f(t)$. In this section, we discretize equation (5.2) using the L1 and L2 method, combining them into one numerical scheme.

Remark 5.20 (Behaviour homogenous vs non homogenous). In the homogeneous setting, is the system's behaviour is dependent on the initial conditions and the interaction between the two memory terms ${}^{C}D^{\alpha}x(t)$ and ${}^{C}D^{\beta}x(t)$. This type of equation models fractional oscillations and will be relevant when we are discussing the harmonic oscillators.

Definition 5.21 (Discrete scheme for multi-term fractional differential equations)**.** Let $x(t)$ be the solution to the two-term Caputo fractional differential equation:

$$ {}_{0}^{C}D_{t}^{\alpha}x(t) + a\ {}_{0}^{C}D_{t}^{\beta}x(t) + b\,x(t) = f(t), \quad t \in [0, T], $$

with initial conditions $x(0) = x_0$ and $x'(0) = x_1$, where $1 < \alpha < 2$, $0 < \beta < 1$, and $a, b \in \mathbb{R}$. The discretization is on a uniform grid $t_n = n\Delta t$, such that the approximation is given by:

$$ W_{-1}x_{n+1} + \sum_{k=0}^{n} W_k x_{n-k} + a \sum_{k=0}^{n-1} b_{n-k-1}\left(x_{k+1} - x_k\right) + bx_n = f(t_n) $$

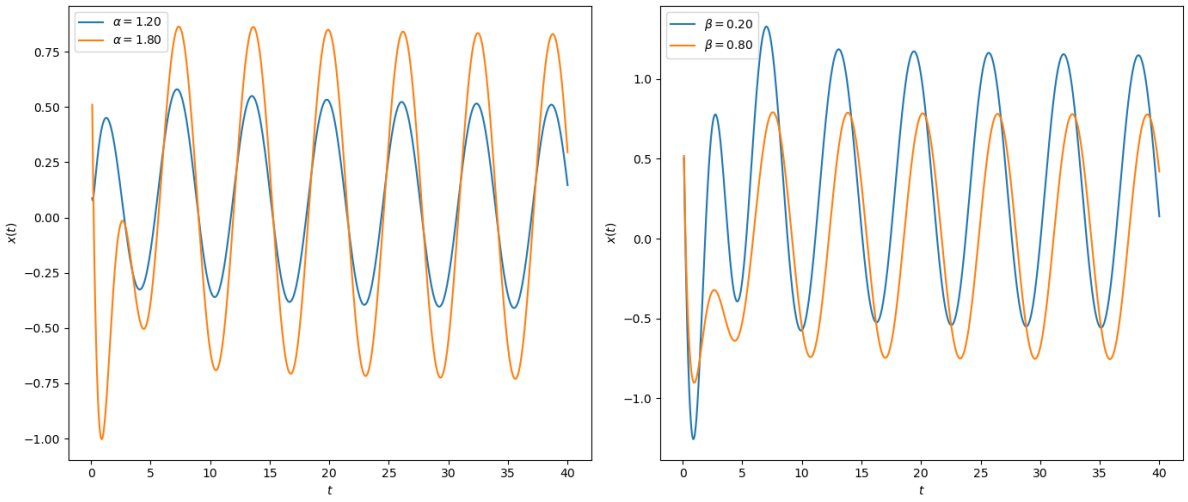


Figure 5.2: Numerical solutions $x(t)$ to the two-term fractional differential equation: ${}_{0}^{C}D_{t}^{\alpha}x(t) + a\ {}_{0}^{C}D_{t}^{\beta}x(t) + b\,x(t) = f(t),, \quad x(0) = x'(0) = 1, \quad a, b = 1,$ using the L2 and L1 schemes. Left: variation in the order of the higher order derivative (α) for a fixed order $\beta = 0.6$. Right: variation in the order of the first derivative (β) for a fixed value $\alpha = 1.8$, where $n = 2000$.

Theorem 5.22 (Error estimate)**.** *Let $x(t)$ be the exact solution to the multi-term Fractional Differential Equation 5.21. Then the error bound of the approximation is given by the bound:*

$$ |x(t) - x(t_n)| \leq C\left(\Delta t^{3-\alpha} + \Delta t^{2-\beta}\right), $$

for some constant $C > 0$ independent of Δt and n.

5.4. Summary and comparison

In this chapter, we discussed the numerical approaches for approximating solutions to fractional differential equations, focusing on the fractional power series (FPS) method and the L1 method for Caputo derivatives.

The FPS method constructs solutions as formal series expansions involving generalized powers and Gamma function coefficients. It is particularly effective when the solution is analytic and has smooth behavior near the initial point. However, its performance becomes worse for larger values of t or near singularities due to truncation and convergence issues.

In contrast, the L1 method is a finite difference scheme that approximates the Caputo derivative using backward differences. Its strength lies in its simplicity and applicability to problems. The L1 method is more well-rounded for larger intervals of t and has predictable error convergence under some assumptions.

Comparatively, the FPS method offers higher theoretical accuracy in smooth settings and can capture local dynamics efficiently with fewer terms. While the L1 method has lower local accuracy, it is more stable, practical, and offers a broader perspective on problems.

In summary, the FPS method is best suited for theoretical validation and small-time behaviour, while the L1 method is more suitable for general numerical simulation and integration of FDEs.

Chapter 6

The fractional harmonic oscillator

In the previous chapters, we built the foundation by analyzing the classical harmonic oscillator and its variants in Chapter 2, followed by the development of fractional calculus in Chapter 3. After that we analytically and numerically solved fractional differential equations in chapters 4 and 5. These chapters laid the groundwork to understand the dynamics of the fractional oscillator that we will discuss in this chapter.

This chapter serves as a bridge between the classical and fractional perspectives. We introduce the fractional harmonic oscillator as a natural extension of its classical counterpart, replacing integer-order derivatives with non-integer orders to observe the effects and dynamics. For each classical case, simple, nonlinear, damped, and driven oscillator we will formulate the corresponding fractional extension and analyze its solution and structure.

6.1. The Simple fractional harmonic oscillator

In the classical setting, the simple harmonic oscillator is built by the second-order differential equation with constant coefficients. This system was introduced in Section 2.1. and serves as a baseline for understanding the behavior. The fractional generalization of this model replaces the second-order derivative with a fractional derivative of order $1 < \alpha \leq 2$.

Definition 6.1 (Fractional simple harmonic oscillator)**.** If the order of the fractional derivative $1 < \alpha \leq 2$ and $\omega > 0$. Then the simple fractional harmonic oscillator is defined by the Caputo fractional differential equation:

$$ {}_0^C D_t^\alpha x(t) + \omega^2 x(t) = 0, \quad x(0) = x_0, \quad x'(0) = x_1. $$

Remark 6.2 (Choice of the Caputo derivative)*.* In this formulation of the system, we use the Caputo derivative due to its compatibility with classical initial conditions. This bridges the classical models in terms of integer-order derivatives. Meaning, that this choice gives a physical interpretation where the initial displacement and velocity are given as conditions.

Theorem 6.3 (Solution of the fractional simple harmonic oscillator)**.** *The unique solution of the simple oscillator is given by:*

$$ x(t) = x_0 E_{\alpha,1}(-\omega^2 t^\alpha) + x_1 t E_{\alpha,2}(-\omega^2 t^\alpha). $$

This result follows directly from the general solution of homogeneous linear fractional differential equations with constant coefficients developed in example 4.2.

Remark 6.4 (Consistency under derivative types). As discussed in Section 3.2.1., the Caputo and Riemann–Liouville derivatives are equivalent when the initial conditions satisfy $x^{(k)}(0) = 0$ for $k = 0, 1, \ldots, \lceil \alpha \rceil - 1$ see theorem 3.27. In this context, the solution coincides with the Riemann-Liouville or Grünwald-Letnikov conditions, giving overarching consistency.

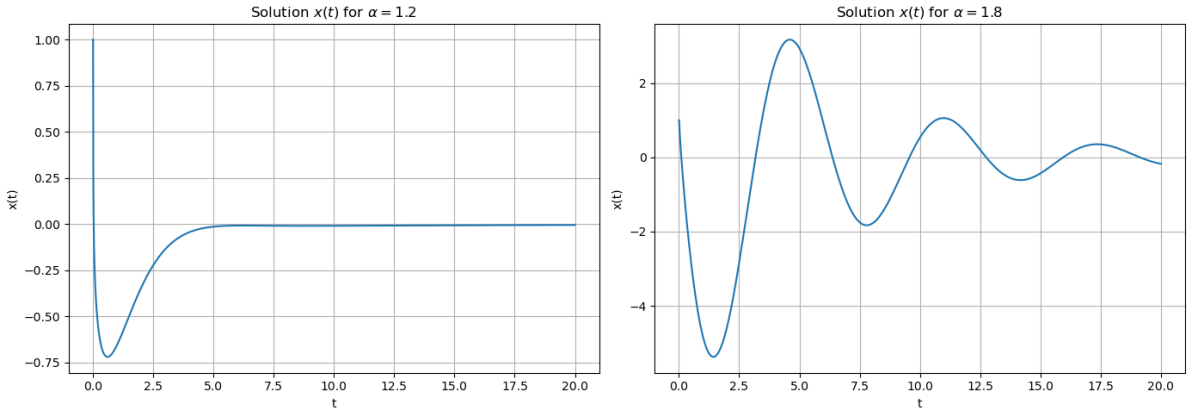


Figure 6.1: Solution curves $x(t)$ for the fractional oscillator with $\omega = 1$, $x_0 = 1$, and $x_1 = 0$ with $n = 2000$ when using L1 and L2. Left: $\alpha = 1.2$ shows slow decaying non-periodical oscillations. Right: $\alpha = 1.8$ closely resembles damped motion with some decay.

Remark 6.5 (Differences in long-term behavior). The solution in theorem 6.3 is composed of two Mittag-Leffler terms, each of which have algebraic decay for negative arguments (see theorem 3.17). This contrasts with the exponential decay of classical damped systems.

Remark 6.6 (Coincide classical and fractional solution). Using theorem 3.31 on the reduction of fractional to classical derivatives, we recover the standard second-order harmonic oscillator in the limit $\alpha \to 2$. The solution then becomes $x(t) = x_0 \cos(\omega t) + \frac{x_1}{\omega} \sin(\omega t)$, which is consistent with lemma 3.14.

Remark 6.7 (Amplitude and phase-shift). Even without an explicit damping term, the system displays some form of dampening, as seen in figure 6.1. This behavior arises from the non-local properties of the Caputo derivative (remark 3.36), where the memory term almost seems like energy leaving the sytem. The more α decreases from 2 the more impact of this effect.

6.1.1. Phase portraits of the simple fractional oscillator

To visualize the system in the phase space, we look at the trajectory $x(t), \frac{x(t)}{dt}$ for different values of the fractional order α. The system:

$$ {}_0^C D_t^\alpha x(t) + \omega^2 x(t) = 0 $$

is solved with initial conditions $x_0 = 1$, $x_1 = 1$, and $\omega = 1$, using the L2 method and $N = 2000$ time steps.

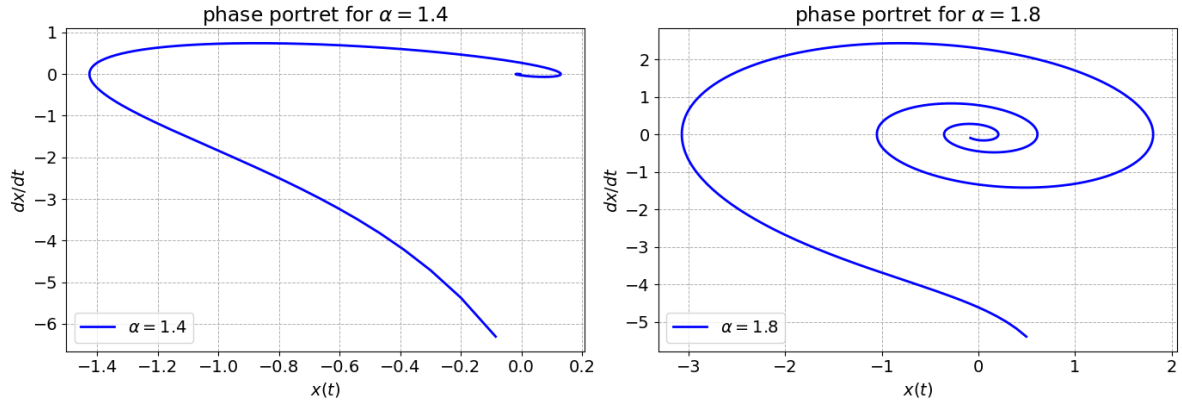


Figure 6.2: Phase portraits $x(t), \frac{x(t)}{dt}$ for the fractional oscillator with $n = 1000$ when using the central finite difference to approximate the derivative of $x(t)$. Left: $\alpha = 1.2$, showing slow inward spiraling due to strong memory effects. Right: $\alpha = 1.8$, exhibiting a nearly classical underdamped phase curve.

Remark 6.8 (Non-classical phase behavior). Fractional phase portraits differ fundamentally from classical ones. When α is close to 2, the system behaves like the harmonic oscillator. For smaller α, memory effects dominates, and the decay makes it that trajectories is spiraling inward more slowly. These phase portraits provide geometric confirmation of the damping and memory properties embedded in the fractional model.

6.2. Non-Linear Fractional Oscillator

The nonlinear oscillators came in to existence when the pendulum's swing was larger with respect to its displacement 2.6. In this section 2.2., we introduced the fractional nonlinear oscillator and its phase portraits.

Definition 6.9 (Fractional Non-Linear Oscillator)**.** If the order of the derivative is $1 < \alpha \leq 2$, then the fractional non-linear oscillator is defined by the Caputo equation:

$$ {}_0^C D_t^\alpha x(t) + \sin(x(t)) = 0, \quad x(0) = x_0, \quad x'(0) = x_1. $$

Remark 6.10 (Existence of solutions)*.* Although the nonlinear fractional oscillator does not admit a exact solution as we had seen in the classical case, this does not exempt the existence and uniqueness of a local solution is guaranteed by theorem 3.47 in Section 3.3.1., since $\sin(x)$ is a smooth function on the interval $[0, T]$.

Remark 6.11 (Lack of closed-form solutions and approach)*.* The nonlinear oscillator does not admit a closed-form solution. Thus our analytical approaches fails, since they rely mostly on the notion of the Mittag-Leffler function, but we can use numerical approximation to analyse this system.

Theorem 6.12 (L2 approximation of the nonlinear fractional oscillator)**.** *We can approximate the non-linear oscillator on the interval $t \in [0, T]$ with the L2 method. Let $\{t_n = n\Delta t\}_{n=0}^N$ be an uniform grid with $\Delta t = T/N$ and define $x_n \approx x(t_n)$. The L2 approximation to the Caputo derivative at time t_{n+1} is given by:*

$$ x_{n+1} = -\sum_{k=0}^{n} W_k x_{n-k} - \sin(x_n). $$

Where the weights W_k are defined in 5.12.

6.2.0.1. Phase portrait of the nonlinear fractional oscillator

Remark 6.13 (Comparison with the linear fractional oscillator)*.* Compared to the linear fractional oscillator:

$$ {}_0^C D_t^\alpha x(t) + \omega^2 x(t) = 0, $$

the nonlinear equation introduces amplitude-dependent frequency and distorted phase geometry. In the linear case, solutions decay algebraically with time but maintain harmonic shape. In contrast, the nonlinear version displays damping rates that depend on the state $x(t)$, as seen in the flattened and compressed phase portraits. This interaction between nonlinearity and memory produces richer dynamics and illustrates how fractional order enhances the complexity of nonlinear systems.

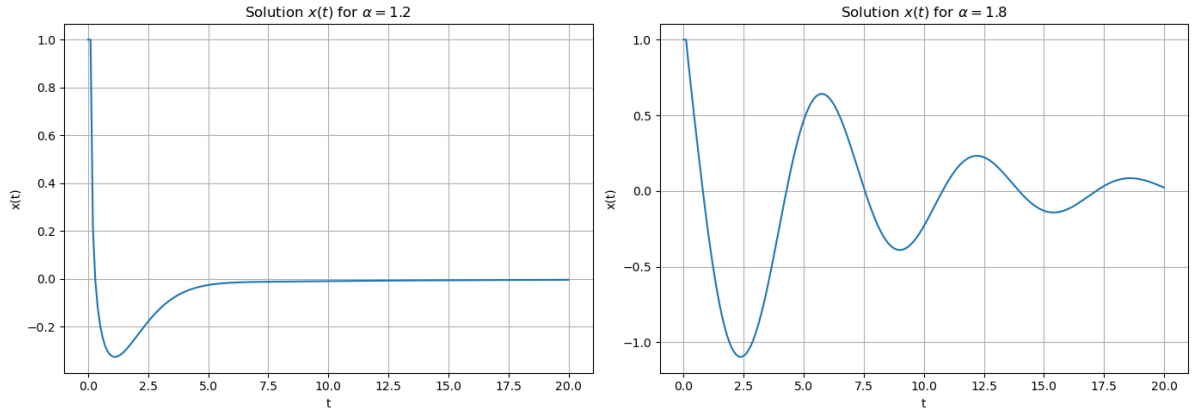


Figure 6.3: Solution curves $x(t)$ of the nonlinear fractional oscillator for different orders α with $n = 2000$. Left: $\alpha = 1.2$, showing strong nonlinear damping. Right: $\alpha = 1.8$, closer to classical oscillations but still influenced by decay.

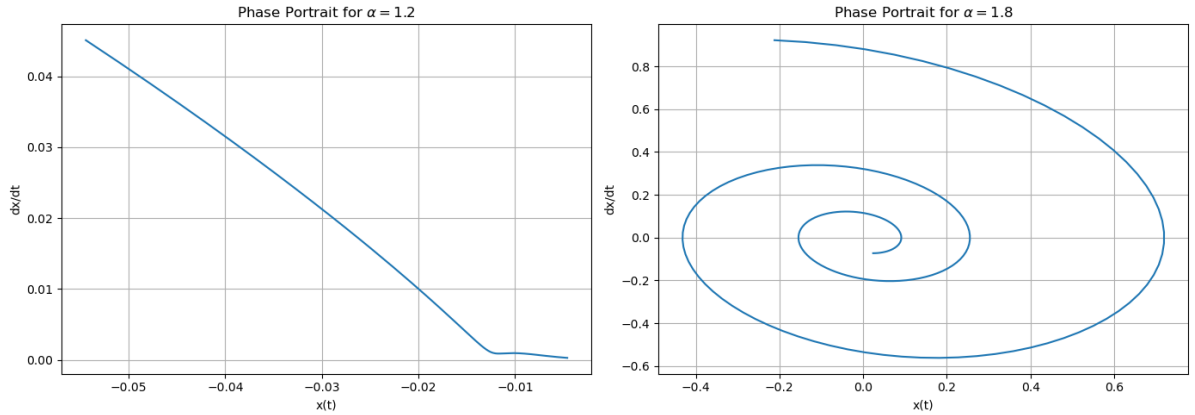


Figure 6.4: Phase portraits $x(t), \frac{x(t)}{dt}$ for the nonlinear fractional oscillator with $n = 1000$ when using the central finite difference to approximate the derivative of $x(t)$. The left panel for $\alpha = 1.2$ shows distorted spiral property. The right panel for $\alpha = 1.8$ resembles a classical nonlinear phase spiral, but with some decay.

6.3. Damped fractional harmonic oscillator

Definition 6.14 (Damped Fractional Oscillator)**.** If the order of the derivatives is $1 < \alpha \leq 2$ and $0 < \beta < 1$, then the damped fractional oscillator is defined by:

$$ {}_0^C D_t^\alpha x(t) + \gamma\, {}_0^C D_t^\beta x(t) + \omega^2 x(t) = 0, \quad x(0) = x_0, \quad x'(0) = x_1. $$

Remark 6.15 (Interpretation of the coefficients)*.* This equation extends the classical damped harmonic oscillator by replacing the second-order and first-order derivatives with their fractional extension. The coefficient ω^2 corresponds to the stiffness constant k in the classical system, and γ plays the role of the damping coefficient b. When $\alpha = 2$ and $\beta = 1$, the system reduces to the classical form.

Theorem 6.16 (Analytical solution of the damped fractional oscillator)**.** *The solution of the Damped Fractional Oscillator is given by:*

$$ {}_0^C D_t^\alpha x(t) + \gamma\, {}_0^C D_t^\beta x(t) + \omega^2 x(t) = 0, \quad x(0) = x_0, \quad x'(0) = x_1, $$

with $1 < \alpha \leq 2$, $0 < \beta < 1$, and $\gamma, \omega > 0$. Then the unique solution is given by the convergent fractional power series:

$$ x(t) = \sum_{k=0}^{\infty} c_k t^{k\gamma}, \quad \textit{where } \gamma = \alpha - \beta, $$

and the coefficients $\{c_k\}$ satisfy the recurrence relation:

$$ c_k = -\frac{1}{\omega^2} \left[\frac{c_{k-1}\, \Gamma(\gamma(k-1)+1)}{\Gamma(\gamma(k-1)+1-\beta)} \cdot \gamma^\beta + \frac{c_{k-2}\, \Gamma(\gamma(k-2)+1)}{\Gamma(\gamma(k-2)+1-\alpha)} \cdot \gamma^\alpha \right], \quad k \geq 2, $$

with initial values $c_0 = x_0$ and $c_1 = x_1/\gamma$ determined by the initial conditions. this solution is the alternative of the differential equation solved in 4.35 , where the power series method gives form to the two-parameter Mittag-Leffler functions.

Remark 6.17 (Fractional versus classical damping)*.* The fractional damped oscillator definition 6.14 generalizes the classical model with two fractional orders. These orders introduce some form of decay rate as seen in 3.17, the phase portrait also support this claim since they are not purely exponential but algebraic.

Remark 6.18 (Fractional damping behaviour)*.*

Example 6.19 (Underdamped, critically damped, overdamped cases for $\alpha = 1.8$, $\beta = 1.0$)**.**

6.3.0.1. Phase portrait of the damped fractional oscillator

To analyze the behavior of solutions of the homogeneous fractional oscillator, we look at the phase space trajectory $x(t), \frac{x(t)}{dt}$ for different values of the fractional orders α and β.

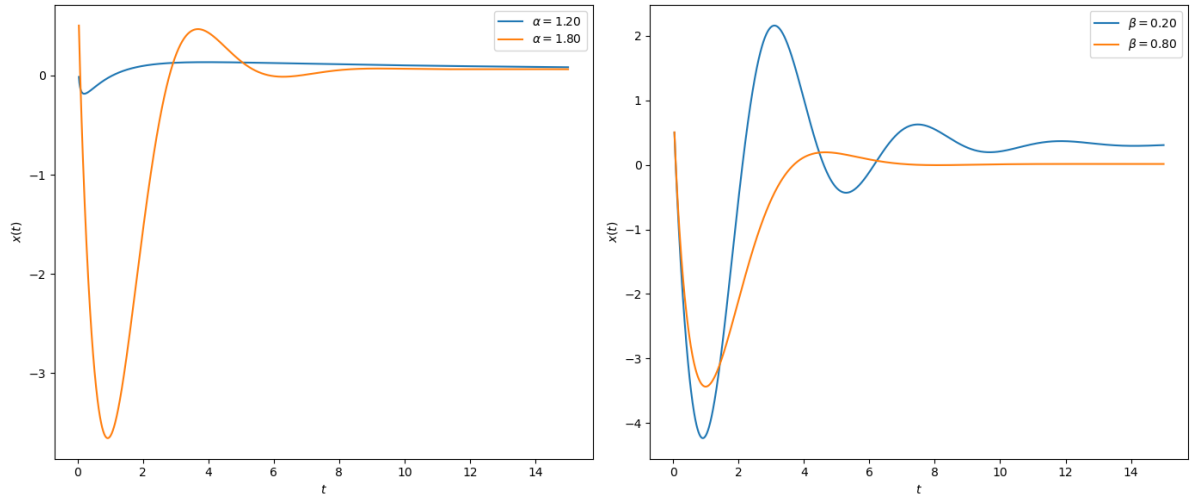


Figure 6.5: Solution $x(t)$ of the homogeneous fractional oscillator with $n = 1000$ for the L1 and L2 method on the interval $[0, 20]$. Left: $\alpha = 1.8$, $\beta = 0.6$, showing damped oscillations with slow algebraic decay. Right: $\alpha = 0.2$, $\beta = 1.2$, rapid overdamped convergence to equilibrium. Both were computed using the L2–L1 scheme with $N = 2000$ steps.

We consider the homogeneous equation

$$ {}_0^C D_t^\alpha x(t) + \gamma \, {}_0^C D_t^\beta x(t) + \omega^2 x(t) = 0, $$

with initial conditions $x(0) = x'(0) = 1$, and fixed parameters $\gamma = \omega^2 = 1$. The system is numerically solved on the interval $[0, 10]$ using $N = 2000$ time steps. The numerical scheme applies the L2 method for the Caputo derivative $D^\alpha x(t)$ (valid for $1 < \alpha < 2$) and the L1 method for the Caputo derivative $D^\beta x(t)$ (used when $0 < \beta < 1$).

Remark 6.20 (Fractional damping without forcing). In the absence of a forcing term, the behavior of the fractional oscillator is overshadowed by the orders α and β and their memory-damping. For $\alpha = 1.8$, $\beta = 0.6$, the solution spirals relative slow toward equilibrium, as it is underdamped (as in the same conditions as 2.9). For $\alpha = 0.2$, $\beta = 1.2$, the phase trajectory decays to the origin fast, reflecting strong damping from the order.

6.4. Driven fractional harmonic oscillator

Definition 6.21 (Driven fractional oscillator)**.** If the order of the derivatives is $1 < \alpha \leq 2$ and $0 < \beta < 1$, then the driven fractional oscillator is defined by the equation,

$$ {}_0^C D_t^\alpha x(t) + \gamma \, {}_0^C D_t^\beta x(t) + \omega^2 x(t) = f(t), \quad x(0) = x_0, \quad x'(0) = x_1. $$

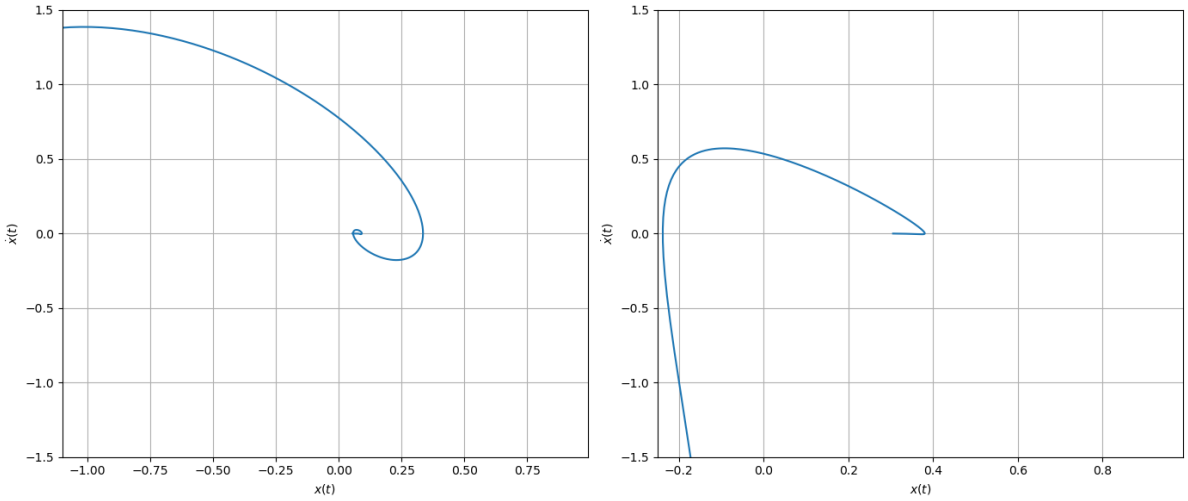


Figure 6.6: Phase portraits of the homogeneous fractional oscillator in two settings with $n = 2000$ when using the central finite difference to approximate the derivative of $x(t)$. Left: $\alpha = 1.8$, $\beta = 0.6$, oscillatory decay. Right: $\alpha = 0.2$, $\beta = 1.2$, showing overdamped effect. Simulations were computed with $N = 2000$ steps using the L2 and L1 schemes for the Caputo derivatives.

Remark 6.22 (Interpretation)*.* If we build on the the previous section on the homogeneous damped fractional oscillator 6.14, the driven fractional oscillator widens the model by adding a forcing term $f(t)$. This term represents an external in force or input, such as a periodic signal.

In the classical case,

$$\frac{d^2x}{dt^2} + \gamma\frac{dx}{dt} + \omega^2 x(t) = f(t)$$

which we discussed in section 2.4.. We had a periodic forcing term $f(t) = F_0\cos(\omega_d t)$, which lead to some chaotic behaviour in the beginning and steadily resonated into a stable state of oscillation. In the fractional setting, the system has some phase delay and memory effects due to the fractional orders α and β. This changes the solution structure and makes it more damped.

Theorem 6.23 (Numerical scheme for the driven fractional oscillator)**.** *Discretizing the Caputo derivatives using the L2 method for α and the L1 method for β on a uniform grid $t_n = n\Delta t$ for $n = 0, 1, \ldots, N$, gives the numerical scheme,*

$$x_{n+1} = f(t_n) - \sum_{k=0}^{n} W_k x_{n-k} - \gamma\sum_{k=0}^{n-1} b_{n-k-1}(x_{k+1} - x_k) - \omega^2 x_n,$$

where W_k are the L2 weights for order α and b_k are the L1 weights for order β as defined in Section 5.8.

To further illustrate the impact of the fractional orders α and β on the solution behaviour, we numerically simulate the two-term damped fractional oscillator

$$D^{\alpha}x(t) + \gamma D^{\beta}x(t) + \omega^2 x(t) = f(t), \quad x(0) = x'(0) = 1,$$

where $f(t) = \cos(t)$, $a = b = 1$, and $x(t)$ is computed using the L2 and L1 methods for orders $1 < \alpha < 2$ and $0 < \beta < 1$, respectively.

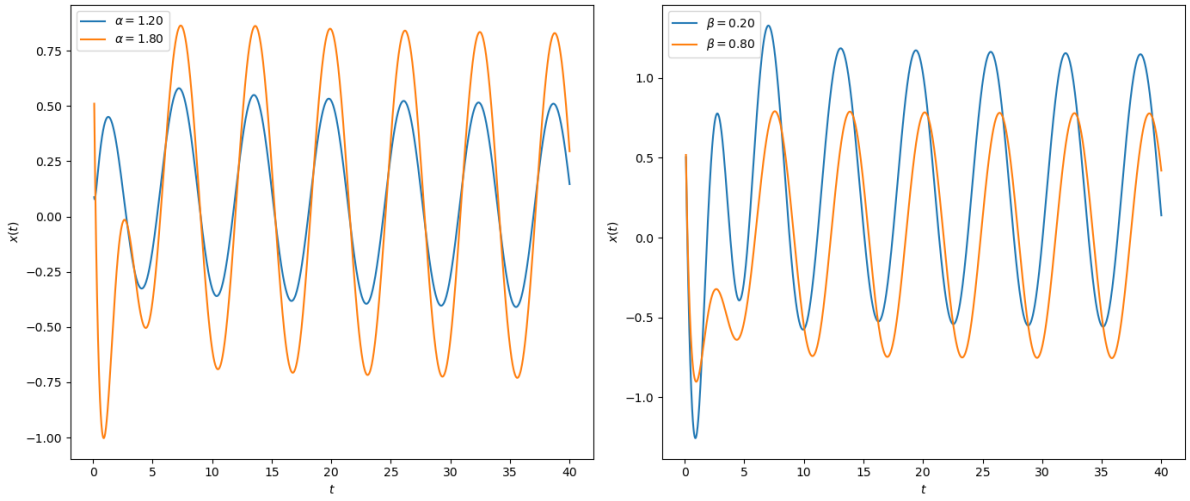


Figure 6.7: Numerical solutions $x(t)$ to the two-term fractional differential equation computed on $[0, 40]$ using $N = 2000$ time steps. Left fig. variates in the higher-order term $\alpha \in \{1.2, 1.8\}$ with fixed $\beta = 0.6$. Right fig. variates in the first-order term $\beta \in \{0.2, 0.8\}$ with fixed $\alpha = 1.8$.

Remark 6.24 (Interpretation of fractional driven oscillator). The two subplots illustrate how the solution $x(t)$ changes with fractional orders α and β. A smaller value of $\alpha = 1.2$ (left-blue fig.) leads to slower memory decay and smoother transitions, whereas $\alpha = 1.8$ (left-yellow fig.) introduces faster decaying system. In the other picture, variation of β affects the damping behaviour, lower $\beta = 0.2$ results in stronger resistance (right-blue fig.), while higher $\beta = 0.8$ produces a faster relaxation toward equilibrium (right-yellow fig.). These behaviours confirm the amping like effect of fractional derivatives 3.36.

Remark 6.25 (Difference in analytical treatment versus numerical scheme). In the classical setting section 2.12, the solution to the driven oscillator is decomposed into a homogeneous solution $x_h(t)$ and a particular solution $x_p(t)$. This allowed us to separately analysis the natural and forced response of the system. However, in the fractional setting, such decomposition is less straightforward due to the non-closed form solution and the numerical framework we take.

Example 6.26 (Response to periodic input)**.**

Remark 6.27 (Driven oscillations and memory effect). The presence of an external force in definition 6.21 introduces a new form of dynamics. If we compare it to

the homogeneous case 6.14, we see that the dampening effect of the memory effect causes the system no longer reaches a pure steady state of oscillation as in the classical case (section 2.4). Instead, it has some memory dampening (see in remark 5.16).

Remark 6.28 (Suppression of resonance)*.* In the classical driven oscillator, resonance occurs when the driving force's frequency matches that of the system (Remark 2.26). In the fractional case, the resonance is suppressed and spread out due to the damping of the memory effect.

Phase portraits of the fractional oscillator

To visualize the behaviour of the fractional driven oscillator, we examine the phase space trajectory $x(t), \frac{x(t)}{dt}$ for different values of the fractional orders α and β. These phase portraits give some insight into how fractional differentiation affects the stability of trajectories in comparison to classical systems.

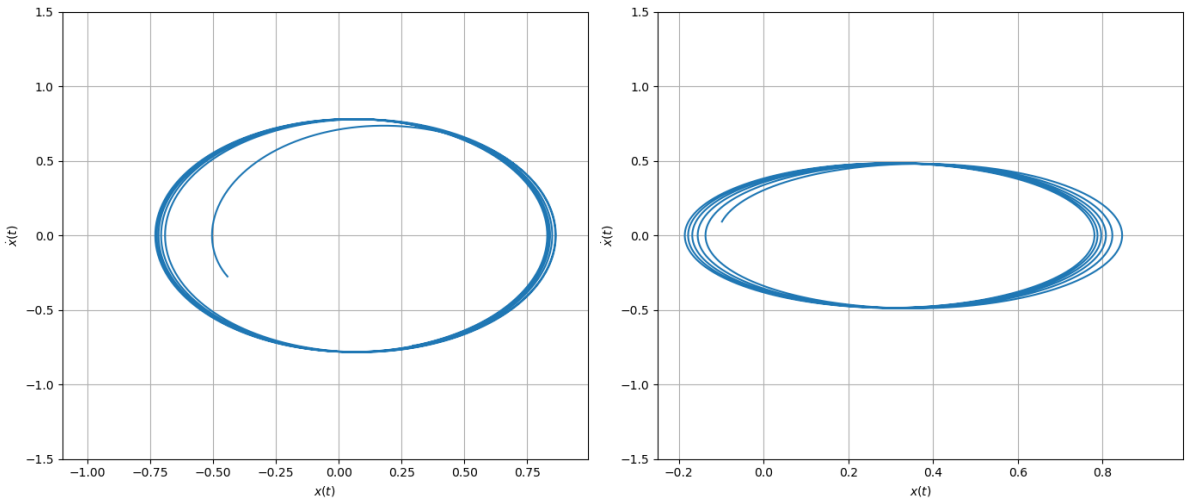


Figure 6.8: Phase portraits of the fractional oscillator for two configurations. Left: $\alpha = 1.8$, $\beta = 0.6$, with moderate damping and memory effect. Right: $\alpha = 0.2$, $\beta = 1.2$, showing stronger damping.

Remark 6.29 (Fractional phase space behaviour)*.* The left portrait in figure 6.8 has slowly decaying oscillations like behaviour, showing memory in the system's velocity since α is close to 2. The spiraling trajectory looks like the underdamped oscillator, but with slower algebraic decay.

In contrast, the right figure with $\alpha = 0.2$, $\beta = 1.2$ shows sharper, damped motion with less oscillation-like behaviour. Here, the damping from $\beta > 1$ suppresses the system oscillatory behavior, and the small α reduces memory effect.

6.5. Summary

This chapter extends classical oscillators to the fractional setting using Caputo derivatives. The simple fractional oscillator (definition 6.1) was solved via Mittag-Leffler functions (theorem 6.3), where the solutions algebraic decay and there is some form of phase delay. The non-linear and damped systems (definitions 6.14 and 6.9) required us to use numerical methods. Phase portraits across cases showed damped spiraling inward behaviour instead of closed orbits. In the driven system (definition 6.21), resonance became weakend and delayed due to fractional damping. Altogether, fractional models have retained their classical structure while adding some non-local damping and memory effects.

Chapter 7

Fractional dynamic systems

7.1. Classical dynamic systems

7.1.1. Van der Pol oscillator

Definition 7.1. (Van der Pol equation)7.2 The Van der Pol oscillator is a non-linear dynamical system given by the second-order differential equation,

$$\frac{d^2x}{dt^2} - \mu(1-x^2)\frac{dx}{dt} + x = 0 \qquad \mu \in \mathbb{R}. \tag{7.1}$$

Here, μ represents the parameter that scales the non-linearity or dampening of the system.

Remark 7.2. When $\mu = 0$ we have that the equation (7.2) reduces to the classical harmonic oscillator,

$$\frac{d^2x}{dt^2} + x = 0,$$

which is known to have closed form solutions.

Example 7.3 (Numerical simulation)**.**

Remark 7.4. For $\mu = 0.5$, the oscillator behaves almost identically to the harmonic oscillator, showing stable oscillations. As μ increases to 2, the non-linear damping becomes more influencial and the waveform begins to sharpen like teeth. At $\mu = 10$, the system displays relaxation oscillations, slow drafts followed by quick transitions.

Lemma 7.5 (System representation)**.** *The Van der Pol oscillator equation can be expressed as a first-order dynamical system in the phase space* (x, y) *with* $y = \frac{dx}{dt}$*,*

$$\begin{cases} \frac{dx}{dt} = y, \\ \frac{dy}{dt} = \mu(1-x^2)y - x. \end{cases} \tag{7.2}$$

Remark 7.6. This representation allows for straightforward numerical analysis and visualization of the phase portraits.

Lemma 7.7 (Equilibrium point)**.** *The Van der Pol oscillator defined has a single equilibrium point at* $(x, y) = (0, 0)$*.*

Theorem 7.8 (Stability of equilibrium)**.** *The equilibrium* $(0, 0)$ *of the Van der Pol oscillator is unstable for all* $\mu > 0$*.*

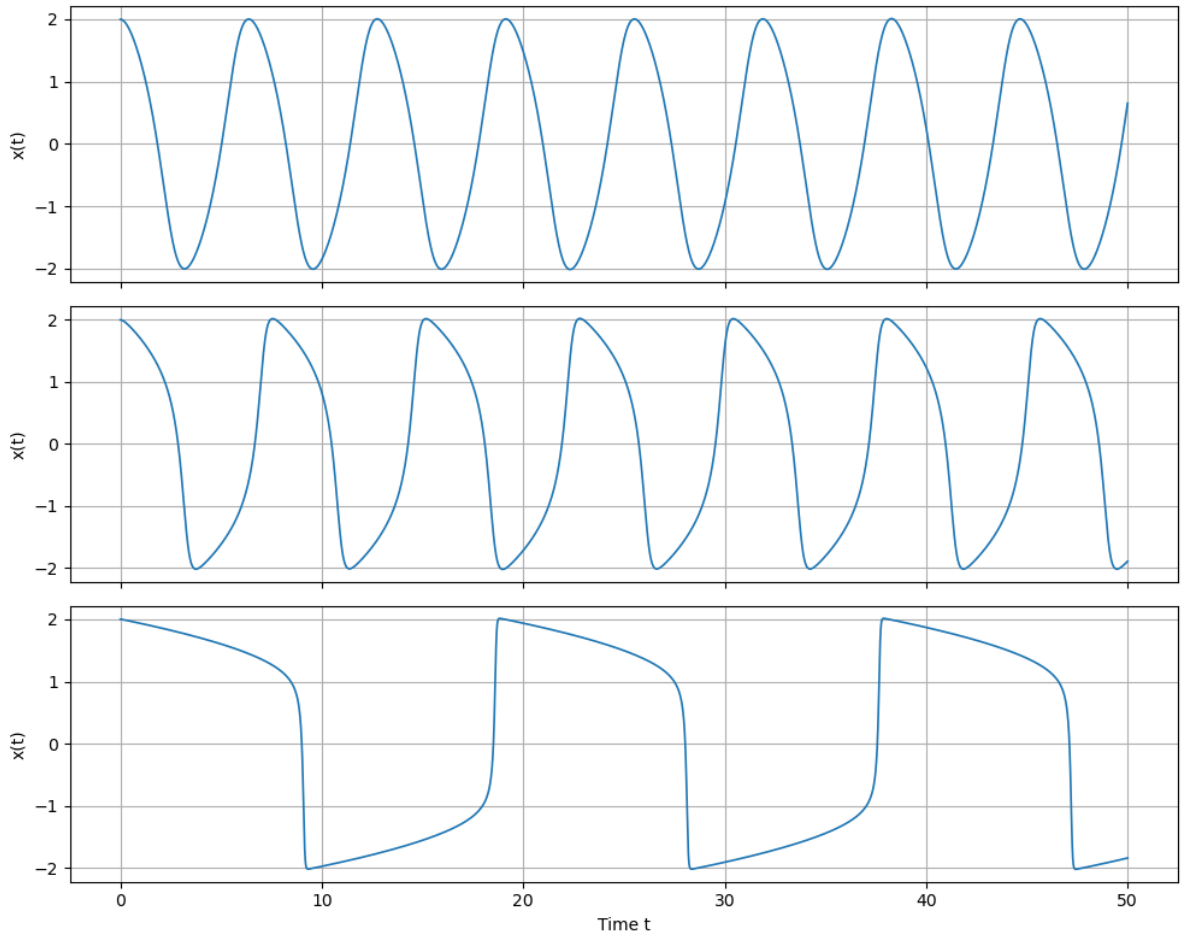


Figure 7.1: The development of $x(t)$ of the Van der Pol oscillator for different values of μ. From top to bottom, $\mu = 0.5$, $\mu = 2$, and $\mu = 10$.

Remark 7.9. Linearizing the system of equations around the equilibrium $(0, 0)$ gives us the Jacobi martrix,

$$J(0,0) = \begin{pmatrix} 0 & 1 \\ -1 & \mu \end{pmatrix}.$$

The eigenvalues λ of the Jacobian are given by $\lambda = \frac{\mu \pm \sqrt{\mu^2 - 4}}{2}$. For $0 < \mu < 2$, are the eigenvalues positive and real therefore unstable spiral. For $\mu \geq 2$, are the eigenvalues real and at least one positive. Thus $(0, 0)$ is always unstable for $\mu > 0$.

Remark 7.10. The existence of a stable limit-cycle is the biggest property of the van der Pol oscillator. For small values of μ we have that the cycles are nearly circular. While for large values of μ the system behaves more like a relaxed oscillator.

Remark 7.11. The amplitude and shape of the limit cycle depend significantly on μ. For larger values of μ, the oscillator shows clear non-harmonic behavior.

Example 7.12 (Numerical simulation)**.**

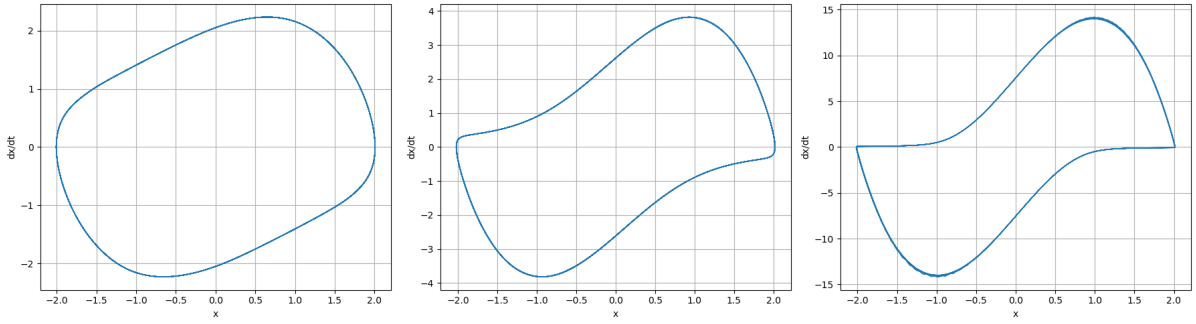


Figure 7.2: Phase plane behavior of the Van der Pol oscillator for various values of μ. Left to right, $\mu = 0.5$, $\mu = 2$, and $\mu = 10$.

7.1.2. Lotka-Volterra equations

Definition 7.13. (Lotka-Volterra system) The Lotka-Volterra equations describe the system given by,

$$\begin{cases} \frac{dx}{dt} &= \alpha x - \beta xy, & \alpha, \beta \in \mathbb{R} \\ \frac{dy}{dt} &= \delta xy - \gamma y, & \delta, \gamma \in \mathbb{R}. \end{cases} \tag{7.3}$$

Remark 7.14. The Lotka-Volterra equations, also known as the predator-prey model, describe the interaction between two species. Where one species is the prey population x and a predator population y over time t. The parameters of the system have a biological interpretations, where α is the natural growth rate of prey, β represents the hunting rate at which predators consume prey, γ is the natural death rate of predators, and δ measures the efficiency of turning consumed prey into predator growth.

Lemma 7.15 (Equilibria of the Lotka-Volterra system)**.** *The Lotka-Volterra system* (7.3) *has two equilibrium points, the trivial equilibrium* $(x, y) = (0, 0)$ *and the so-called "coexistence" equilibrium* $\left(\frac{\gamma}{\delta}, \frac{\alpha}{\beta}\right)$.

Theorem 7.16 (Stability analysis of equilibria)**.** *The trivial equilibrium* $(0, 0)$ *is a saddle point and therefore unstable. The equilibrium* $\left(\frac{\gamma}{\delta}, \frac{\alpha}{\beta}\right)$ *is a centre and has neutrally stable oscillating behaviour.*

Remark 7.17. The periodic solutions around the co-existence equilibrium form closed orbits. Which tells us that there is a stable predator-prey cycles.

Example 7.18 (Numerical Simulation)**.** Typical behavior of predator-prey cycles can be demonstrated through numerical simulation.

The numerical simulation shows periodic oscillations of the population sizes cycling around the "coexistence" equilibrium.

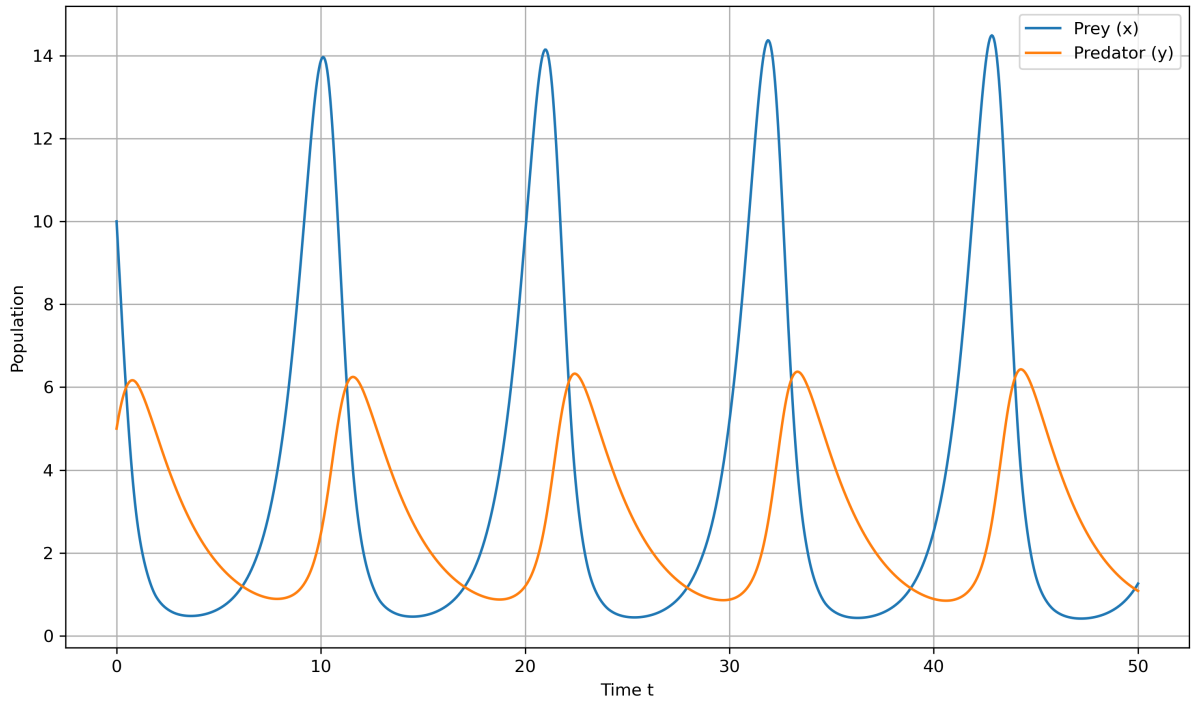


Figure 7.3: The development of the Lotka-Volterra system. The top plot shows the prey population $x(t)$ and the bottom plot shows the predator population $y(t)$. The parameters used are $\alpha = 1.1$, $\beta = 0.4$, $\delta = 0.1$, and $\gamma = 0.4$.

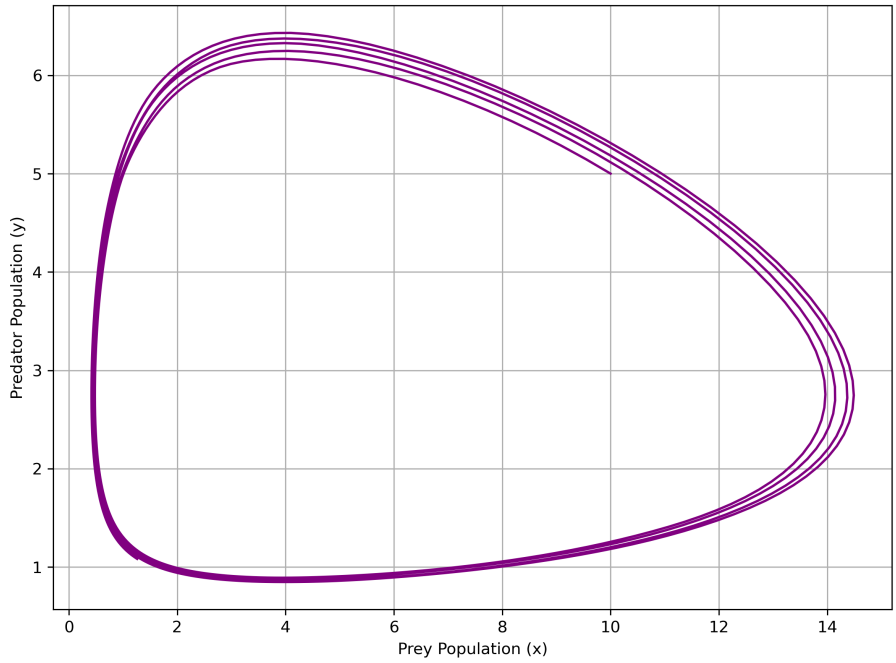


Figure 7.4: Phase portrait of the Lotka-Volterra predator-prey model. The trajectory shows the dynamics between prey and predator populations in the phase plane.The parameters used are $\alpha = 1.1$, $\beta = 0.4$, $\delta = 0.1$, and $\gamma = 0.4$.

Remark 7.19 (Outlook from classical to fractional systems)*.* The Van der Pol and Lotka–Volterra equations are standard examples of non-linear dynamical systems. Which reveal complex behaviour like limit cycles and predator-prey oscillation.

When placed in the broader context of this thesis, these systems complete the basis on which we begon. Although the construction of fractional dynamic systems was not analyzed, the formulation of their classical counterparts in this section marks a natural conclusion to the goal of this thesis. Future exploration, as outlined in Chapter 8.2., will build on this point to further develop the theory and applications of fractional dynamical systems.

Chapter 8

Conclusion

8.1. Summary of findings

This thesis analysed the fractional harmonic oscillator by extending classical models with fractional derivatives to capture memory and non-local effects. Chapter 2 introduced classical oscillators as a foundation. Chapter 3 formalized fractional calculus, highlighting the Caputo derivative for physical and numerical modelling, the role of the Gamma and Mittag-Leffler functions, and the operational framework for FDEs. Chapter 4 developed analytical methods using Mittag-Leffler eigenfunctions, Laplace transforms, and fractional power series to solve linear and some non-linear FDEs. Chapter 5 presented numerical schemes L1 and L2 for the Caputo derivatives. Chapter 6 applied these tools to fractional versions of the simple, damped, driven, and non-linear oscillators, revealing memory damping, algebraic decay, and phase shifting. Chapter 7 concluded with classical dynamical systems, showing how their structure has an extension to future work in the fractional study.

8.2. Future work

This thesis has focused on the fractional harmonic oscillator, both as an analytical object and as a few case studies in fractional modelling and numerical simulation. However, it also reveals a broader and largely unexplored direction, the extension of non-linear dynamical systems to the fractional setting, like the Van der Pol and Lotka-Volterra equations.

While the classical harmonic oscillator is linear and relatively well behaved, many real-world systems exhibit non-linearity, self-interaction, and other forms of feedback properties that are important to models like the van der Pol oscillator or predator-prey systems. Extending these classical models to include fractional-order derivatives would allow us to study how memory effects modify known behaviours such as damping, limit cycles, bifurcations, or even chaos.

The groundwork laid in this thesis helps in the construction and analysis of numerical methods such as the L1 and L2 schemes that can be applied to these more complex systems. A future research direction would be to:

- Develop and analyze fractional generalizations of classical non-linear systems;

- Investigate whether classical phenomena such as bifurcations or attractors stay, shift, or vanish under fractional dynamics;
- Extend the numerical schemes used here to simulate multi-variable and fractional systems;
- Compare fractional models to empirical data in systems where memory effects are known to play a role.

Although this thesis does not include a chapter on fractional dynamical systems, the results obtained for the fractional harmonic oscillator suggest that such an extension is both natural and necessary. With the tools in place, the next step is to apply them to the models to deepen our understanding of fractional calculus, but also look at the broader application to problems in physics, biology, and engineering."

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